

Bank Competition, Defaulting Risks and Margins of Innovation

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Abstract. This paper studies how bank competition affects innovation when firms differ in default risk and banks cannot observe project risk perfectly. We develop a model in which banks compete in capacities and then loan rates while allocating credit between low- and high-default-risk firms. Stronger bank competition lowers loan rates for low-default-risk firms and expands innovation participation on the extensive margin, but relaxes credit rationing for high-default-risk firms and raises innovation investment on the intensive margin. The model also predicts that aggregate loan risk may rise even when loan quality weakly improves within each borrower segment, because competition reallocates lending toward intrinsically riskier firms. Using the deregulation of city commercial banks in China as plausibly exogenous variation in bank competition, we find that increased competition raises R&D participation for state-owned firms and R&D intensity for private firms conditional on innovating. Counterfactual analysis shows that aggregate non-performing loans can be hump-shaped in competition and that removing loan guarantees for state-owned firms reallocates credit toward private firms, raising aggregate R&D output but reducing bank and state-owned-firm welfare.

Résumé.

JEL classification: G21, D22, D82, L13, O31

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1. Introduction

Banks remain the dominant external financiers of firms in many developing economies, so changes in banking competition can materially affect which firms innovate and how much they invest in R&D.¹ Yet the effect of competition is theoretically ambiguous. More competition can lower loan rates and expand credit, but it can also reshape the composition of borrowers and alter the extent of credit rationing under asymmetric information. For innovation finance, this ambiguity is especially important because innovative projects are risky, difficult to verify, and often rely heavily on bank credit in settings where capital markets are less developed. This paper asks a simple question with first-order policy relevance: when firms differ in default risk, how does stronger bank competition change the extensive margin of innovation participation, the intensive margin of innovation spending, and the riskiness of bank loan portfolios?

Our starting point is that competition need not affect all firms in the same way. In China, state-owned firms (SOFs) and private firms (PFs) face systematically different credit conditions. SOFs benefit from relationship lending and implicit government support, so banks treat them as lower-default-risk borrowers. PFs, by contrast, are more exposed to repayment risk and more likely to face credit rationing (Cull and Xu 2003, Ge and Qiu 2007, Guariglia et al. 2011). This institutional setting implies that competition can operate through two distinct margins. For firms that can already obtain credit, stronger competition mainly lowers loan rates and changes participation in innovation. For firms that are rationed, stronger competition can instead relax quantity constraints and increase the scale of innovative investment conditional on participating. That distinction between extensive and intensive margins is central to our analysis.

We formalize this idea in an IO-banking model with heterogeneous borrowers and asymmetric information. Entrepreneurs differ in project risk and in default risk. Banks compete in a two-stage oligopoly, choosing loan capacities and then loan rates, and they cannot observe each borrower's project risk. Building on the credit-rationing logic of Stiglitz and Weiss (1981), the model yields a sharp asymmetric prediction. Low-default-risk firms are fully funded in equilibrium, so more bank competition works through prices and increases innovation on the extensive margin by lowering the threshold for profitable borrowing. High-default-risk firms, by contrast, are credit-rationed, so more competition raises innovation on the intensive margin by increasing the loan size available to active borrowers. We establish existence and uniqueness of the equilibrium and derive comparative statics of increasing

¹ See Cole and Wolken (1995), Benfratello et al. (2008), Robb and Robinson (2014) for further discussion on the role of banks in developing economies.

bank competition on firms' R&D investment margins, loan allocation, loan default risk, and R&D output.

The model also delivers a broader insight about competition and financial stability. A standard concern in the related literature is that competition can worsen loan quality by weakening screening or intensifying adverse selection. Our mechanism is different. In our framework, stronger competition can improve or leave unchanged loan quality within each borrower segment, yet still increase aggregate non-performing loans because banks reallocate credit toward private firms, which are intrinsically riskier. Aggregate portfolio risk therefore depends not only on within-segment loan quality, but also on how competition shifts credit across segments. This portfolio-composition channel is one of the paper's central economic mechanisms.

We then take the model to Chinese data. We exploit the deregulation that expanded city commercial banks as a source of plausibly exogenous variation in local banking competition and combine city-level banking data with firm-level manufacturing data. The empirical results closely track the theory. More competition increases the probability of positive R&D investment for SOFs, but has little effect on their conditional R&D intensity. For PFs, the main effect runs in the opposite margin: banking competition raises R&D intensity conditional on innovating, with much weaker effects on the extensive margin. These patterns are robust across alternative measures of competition, different ownership classifications, and additional specifications.

Finally, we calibrate the model to match key moments from the Chinese data and use it for counterfactual analysis. The quantitative results show that the relationship between bank competition and aggregate loan risk can be hump-shaped once the private-firm market becomes market-clearing. They also show that removing loan guarantees for SOFs reallocates credit toward PFs, raises aggregate R&D output, and reduces the welfare of banks and SOFs. These counterfactuals sharpen the paper's substantive message: policies that protect safer but less productive borrowers may stabilize part of the banking system while distorting the allocation of innovative credit.

The paper contributes in three ways. First, it connects the innovation-finance literature to the IO-banking literature by embedding heterogeneous innovative firms into a model of bank competition with adverse selection, capacity choice, and segmented lending. Second, it shows that decomposing innovation into extensive and intensive margins is not just empirical book-keeping: the two margins correspond to distinct credit-market mechanisms and have different implications for aggregate loan risk. Third, it provides evidence from China that the same increase in bank competition can expand innovation participation for low-default-risk firms and innovation intensity for high-default-risk firms, consistent with the model's mechanism.

The rest of the paper proceeds as follows. Section 2 positions the paper in the related literature. Section 3 presents the model and derives the main

mechanisms. Section 4 brings the model to the data. Section 5 calibrates the model and studies the counterfactuals. Section 6 concludes.

2. Related Literature

This paper is primarily related to the IO-banking literature on competition under asymmetric information. The classic credit-market literature shows that equilibrium credit allocation can be distorted even when borrowers are willing to pay higher loan rates (Akerlof 1970, Rothschild and Stiglitz 1976, Stiglitz and Weiss 1981). Building on that insight, a large IO-banking literature studies how bank competition affects lending outcomes when banks face adverse selection, informational frictions, or endogenous screening. Broecker (1990) shows that, with independent creditworthiness tests, more competition can lower average borrower quality through a winner's-curse effect. Marquez (2002), the paper closest to ours, studies a model with lending-capacity constraints and proprietary information, and shows that stronger competition can disperse information across banks and reduce loan quality. Gehrig (1998) emphasizes that stronger competition may weaken screening incentives and deteriorate loan portfolio quality, while Riordan (1993), Sharpe (1990), Dell'Ariccia et al. (1999) and Von Thadden (2004) analyze how market structure interacts with private information and lending rents more generally.

Our paper contributes to this literature by emphasizing a different mechanism linking bank competition to lending outcomes and financial stability. We do not rely on weaker screening, endogenous learning, or a deterioration in information quality to generate changes in aggregate loan performance. Instead, banks allocate credit across segmented borrower groups that differ in baseline default risk and default protection. As competition intensifies, loan quality can improve or remain unchanged within each segment, yet aggregate non-performing loans can still rise because lending is reallocated toward intrinsically riskier borrowers. The contribution of the paper to the literature is therefore to highlight a portfolio-composition channel through which competition affects aggregate loan risk in a multi-market lending environment.

This perspective also helps reconcile competing views of bank competition. The funds-availability view emphasizes lower borrowing costs and expanded credit under stronger competition, whereas the asymmetric-information view emphasizes persistent rationing and the possibility that competition worsens lending outcomes (Rice and Strahan 2010, Lester et al. 2019, Mahoney and Weyl 2017). Our model combines these forces rather than treating them as alternatives. Competition lowers rates and expands participation for one borrower segment, but relaxes individual loan size for another. The relevant issue is therefore not whether competition is uniformly beneficial, but how competition changes the allocation of credit across borrower types and across lending margins.

The paper also speaks to the literature on banking competition and innovation, but more selectively. Existing empirical work shows that more competitive banking systems can stimulate innovation, especially among firms that rely heavily on external finance (Benfratello et al. 2008, Chava et al. 2013, Cornaggia et al. 2015). Our contribution is to show that this response should be decomposed into extensive and intensive margins, because those margins correspond to distinct credit-market mechanisms. A lower loan rate can induce additional firms to undertake R&D, whereas a relaxation of credit rationing can increase R&D spending conditional on innovating. This distinction is central to both our theory and our empirical design.

Finally, the China setting gives economic content to the model's borrower segmentation without requiring a long institutional detour. A large literature shows that SOFs and PFs differ systematically in credit access, political support, and innovation incentives (Brandt and Li 2003, Ge and Qiu 2007, Guariglia et al. 2011, Firth et al. 2009, Hu and Jefferson 2009, Choi et al. 2011, Aghion et al. 2013). We use these differences in a focused way: SOFs map naturally to lower-default-risk firms because of implicit protection, while PFs map to higher-default-risk firms because they face tighter credit constraints and bear default losses more fully. This makes China a useful setting for testing a mechanism that is fundamentally about competition, credit allocation, and borrower heterogeneity.

3. Theory

This section develops a model to study how stronger bank competition changes credit allocation when firms differ in default risk and banks face asymmetric information about project risk. The main finding is that competition need not move all borrowers along the same margin: when low-risk firms are already fully funded, competition works mainly by lowering loan rates and expanding participation; when high-risk firms are rationed, competition works mainly by relaxing quantity constraints and raising loan size conditional on borrowing.

3.1. Entrepreneurs

The economy consists of a unit continuum of risk-neutral entrepreneurs, each endowed with one unit of wealth and a risky R&D project requiring investment. Entrepreneurs can invest their initial wealth in a safe asset with a constant rate of return, but this wealth cannot directly finance their R&D project. To fund their projects, entrepreneurs seek bank loans, using their initial wealth as collateral. Their objective is to maximize the expected payoff at the end of the period.

Entrepreneurs are heterogeneous in the riskiness of their R&D projects and loan default risks. Each entrepreneur is characterized by a vector (o, θ_o) , where $\theta_o \in [\underline{\theta}, \bar{\theta}] \subset [0, 1]$ represents the riskiness of their R&D project, and $o \in \{l, h\}$ indicates the type of loan default risk, i.e., low default risk (LDRFs,

type- l) or high default risk (HDRFs, type- h). The mass of HDRFs is λ_h , and the mass of LDRFs is $\lambda_l = 1 - \lambda_h$. For each type- o entrepreneur, θ_o follows a known distribution with CDF $F_o(\theta)$ and PDF $f_o(\theta)$. We assume $f_o(\theta)$ is continuously differentiable, non-zero, and bounded on $[\underline{\theta}, \bar{\theta}]$. Moreover, the return on the safe asset for type- o entrepreneurs is b_o .

An entrepreneur of type (o, θ_o) succeeds in the R&D project with probability $1 - \theta_o$ and fails with probability θ_o . With an investment of z units, a successful project generates output $y = \frac{\mu_o \min\{z, \bar{z}_o\}}{1 - \theta_o}$, where $\bar{z}_o$ is the maximum investment allowed by the R&D technology. A failed project produces no output. Thus, while type- o entrepreneurs' R&D projects have the same expected output of $\mu_o z$, they differ in output risk, with higher θ_o indicating greater volatility.

If a type- l entrepreneur defaults on a bank loan, a third party (e.g., the government) may provide a subsidy with probability δ , covering the difference between the loan repayment and the R&D output. In such cases, the third party assumes control of the loan collateral. Thus, δ influences the bank's expected returns on loans to type- l entrepreneurs, but does not affect the entrepreneur's default decision. The parameter δ captures the institutional feature that, in China, SOFs may receive implicit or explicit government support when facing financial distress. In practice, this support can take the form of fiscal transfers, policy-directed restructuring, or other coordinated arrangements that effectively enable lending banks to recover the full contractual repayment despite project failure.

We assume that the type o is observable by all agents, but θ_o is the entrepreneur's private information. While banks know the distribution of θ_o for each type o , they cannot observe an individual entrepreneur's specific θ_o .

A loan contract for R&D investment is characterized by a tuple $\sigma_o \equiv \{r_o, \kappa_o, \hat{z}_o\}$, where r_o is the gross loan interest rate, κ_o is the collateral ratio (the fraction of the loan entrepreneurs must secure with their own wealth), and $\hat{z}_o \in [0, \bar{z}_o]$ is the maximum loan amount an entrepreneur can obtain (credit limit). When an entrepreneur decides to invest in R&D, they apply to a bank for a loan of size z , subject to the collateral constraint $\kappa_o z \leq 1$ and the credit limit $z \leq \hat{z}_o$. Since the entrepreneur cannot directly invest their initial wealth in the R&D project, it is invested in a safe asset, yielding a gross return of b_o . If the R&D project succeeds, the entrepreneur repays the loan; otherwise, they default, and the bank retains the collateral, the interest on the safe asset borne by the collateral, and the residual value of the project.²

² In addition to adverse selection, a moral hazard problem may also arise (e.g., [Bester and Hellwig 1987](#); [Tirole 2006](#)). Specifically, LDRFs and HDRFs can endogenously choose the level of riskiness of their R&D projects. We have extended the model to incorporate a setting where entrepreneurs within both LDRFs and HDRFs can endogenously select the riskiness of their projects after the loan contract is signed. In this framework, higher loan interest rates incentivize entrepreneurs to undertake riskier projects, which in turn reduces banks' expected returns. However, we show that the

Thus, the expected profit for a type- (o, θ_o) entrepreneur from a loan contract is:

$$v_o(\theta_o, \sigma_o) = \max_{z \in [0, \min\{\bar{z}_o, \hat{z}_o, 1/\kappa_o\}]} \{\mu_o z - (1 - \theta) r_o z - \theta b_o \kappa_o z\}. \quad (1)$$

Throughout the analysis, we focus on loan contracts with $r_o > b_o \kappa_o$, as entrepreneurs would otherwise repay the loan even if the R&D project fails. We assume $\mu_o > b_o \kappa_o$ for each $o \in \{h, l\}$ to ensure that R&D projects are profitable, at least for defaulting entrepreneurs. An entrepreneur will choose to invest in R&D if and only if their riskiness exceeds a certain threshold, i.e.,

$$\theta_o \geq \hat{\theta}_o \equiv 1 - \frac{\mu_o - b_o \kappa_o}{r_o - b_o \kappa_o}. \quad (2)$$

In this case, the optimal loan size is $\min\{\bar{z}_o, \hat{z}_o, 1/\kappa_o\}$. The following lemma shows that if the expected R&D output exceeds the collateral value, the threshold $\hat{\theta}_o$ increases with the loan interest rate.

LEMMA 1. *If $\mu_o > b_o \kappa_o$, then $\hat{\theta}_o$ increases in r_o .*

Proof. Eq. (2) implies $\frac{d\hat{\theta}_o}{dr_o} = \frac{\mu_o - b_o \kappa_o}{(r_o - b_o \kappa_o)^2}$, which is positive if and only if $\mu_o > b_o \kappa_o$. ■

To ensure the feasibility of loan contracts, we focus on loan contracts where the collateral ratio κ_o lies within the interval $[\underline{\kappa}, \bar{\kappa}] \subset [0, \mu_o/b_o]$, and the loan interest rate $r_o \leq r_o^{\max}$. Here, $r_o^{\max}$ is the rate corresponding to $\hat{\theta}_o = \bar{\theta}$. We also denote $r_o^{\min}$ as the interest rate corresponding to the threshold $\hat{\theta}_o = \underline{\theta}$. For any $r_o \leq r_o^{\min}$, we set $\hat{\theta}_o = \underline{\theta}$ as all entrepreneurs are willing to do R&D.

3.2. Banks

The banking sector consists of N identical and oligopolistic banks, indexed by $i \in \mathcal{N} \equiv \{1, 2, \dots, N\}$, where N is fixed and exogenously given. As we will show later, N reflects the intensity of competition in the banking sector, with a larger N corresponding to higher competition. Banks provide loans to HDRFs and LDRFs, in each loan market, they engage in price competition subject to individual capacity constraints specified below.

Competition among banks is modeled as a two-stage game. In the first stage, each bank i independently chooses capacity levels for HDRF loans $L_{h,i}$, LDRF loans $L_{l,i}$, and reserves R_i . Loan capacities represent the maximum quantities of loans the bank can supply in the corresponding markets. Reserves are riskless assets that yield an exogenous gross return ρ_r . To manage these

qualitative results of the model remain consistent with the benchmark model. This is because the moral hazard problem affects LDRFs and HDRFs similarly, so adverse selection remains the dominant issue. All results are available upon request.

assets, banks incur costs $c_r(R_i)$, $c_h(L_{h,i})$, and $c_l(L_{l,i})$, which are strictly increasing, strictly convex, and differentiable, with $c_o(0) = 0$ for $o \in \{h, l, r\}$. The reserve cost $c_r(R_i)$ captures the balance sheet cost of holding reserves, while the loan costs $c_h(L_{h,i})$ and $c_l(L_{l,i})$ represent capacity installation costs in the loan markets and reflect the operational expenses associated with loan provision. The two-stage structure reflects the idea that banks must first invest in observable “capacity” (e.g., branches, ATMs, and other service facilities) before competing for customers through price setting in the second stage.

Banks fund their assets through deposits. The deposit supply to each bank is given by $D(N)$ at a regulated deposit rate $\bar{r}_d$, where $D(N) = \bar{D}_d/N$ decreases in N , and $\bar{D}_d$ denotes aggregate deposit supply. This formulation captures institutional features of the Chinese banking system, where deposit rates are subject to ceilings imposed by the central bank and deposit allocation depends on bank size and market structure. Although this deposit market outcome can be microfounded by a perfectly competitive deposit market with binding interest rate regulation, we abstract from these details in the main text to maintain focus on loan market competition, which constitutes the core mechanism of the model. The deposit-market microfoundation is therefore relegated to Online Appendix B.1.

In the second stage, banks’ capacities are fixed at the levels chosen in the first stage, and the vector of capacity choices $\{L_{h,i}, L_{l,i}\}_{i=1}^N$ is common knowledge. At the beginning of this stage, banks independently post loan contracts $\{\sigma_{h,i}, \sigma_{l,i}\}$ for HDRFs and LDRFs, respectively. In this contract, we assume the loan rate and credit limit are endogenously chosen by banks, but the collateral ratio is fixed and equal to κ_o . We refer to this second-stage game as the *price subgame*. Each bank anticipates facing a residual demand function that depends on its own price and capacity relative to those of its competitors. Banks can costlessly supply any amount of loans or deposits up to their installed capacity in each market, but cannot exceed these limits. Consequently, a bank’s actual sales are the minimum of its residual demand and its capacity. Each bank maximizes expected profit, defined as second-stage revenue net of the capacity costs incurred in the first stage.³

Entrepreneurs observe the loan contracts and make R&D decisions. For those who choose to undertake R&D, they simultaneously make loan applications for the banks they want to borrow from. The loan applications are costless. We consider random rationing, i.e., a bank’s loans are sold randomly to the borrowers who submit applications to the bank. The borrowers left for the higher-priced banks would then be a random sample of the borrower population. If the entrepreneur is financed, she devotes all the loan funds to R&D, and we assume there is no moral hazard. Otherwise, the firm exits the

3 We abstract from endogenous screening or learning by lending. Borrower default risk is observable to banks at the time of contracting, and banks differ in lending outcomes solely due to differences in borrower types and policy-induced default protection.

market and earns the returns from safe assets. Moreover, for a loan of size z , the bank incurs a cost of γz to assess the collateral in the event of default. This cost γz represents the bank's transaction costs associated with taking possession of and liquidating the collateral, as discussed in Barro (1976), or the monitoring costs required to verify the borrower's R&D output in the case of default, as explained in Williamson (1987).

Now we define and characterize the properties of loan returns that will be used for the equilibrium analysis. For notational simplicity, we denote $G_o(x) \equiv \mathbb{E}[\theta_o | \theta_o \geq x]$ as the conditional mean for θ_o greater than x , for both $o \in \{h, l\}$.

The default risk and collateral valuation cost imply that the realized loan return is lower than the contractual loan rate. For HDRF loans, the unit average return is given by

$$\begin{aligned} \bar{\rho}_h(r_h) &= \frac{\int_{\hat{\theta}_h(r_h)}^{\bar{\theta}} [(1-\theta)r_h + \theta(b_h\kappa_h - \gamma)] dF_h(\theta)}{1 - F_h(\hat{\theta}_h(r_h))} \\ &= r_h - \mathbb{E}[\theta | \theta \geq \hat{\theta}_h(r_h)] [r_h - (b_h\kappa_h - \gamma)], \end{aligned} \quad (3)$$

where $\mathbb{E}[\theta | \theta \geq \hat{\theta}_h(r_h)]$ is the expected default rate, and $r_h - (b_h\kappa_h - \gamma)$ denotes the unit loss conditional on default. Let $\hat{r}_h$ be the loan rate that maximizes the bank's return $\bar{\rho}_h(r_h)$, i.e., the return-maximizing rate. We focus on the case in which $\hat{r}_h$ is unique and interior. The following lemma provides sufficient conditions for the existence of such a rate:

LEMMA 2. Suppose (i) $G_h''(x) \geq \frac{2}{1-x}G_h'(x)$; (ii) $f_h(\underline{\theta}) \left[1 + \frac{(1-\underline{\theta})\gamma}{\mu_h - b_h\kappa_h}\right] < \frac{1 - \mathbb{E}[\theta_h]}{(1-\underline{\theta})(\mathbb{E}[\theta_h] - \underline{\theta})}$; (iii) $\gamma > \frac{\mu_h - b_h\kappa_h}{1-\underline{\theta}}$. The return-maximizing rate $\hat{r}_h$ is unique and satisfies $\hat{r}_h \in (r_h^{\min}, r_h^{\max})$.

Condition (i) in Lemma 2 guarantees the uniqueness of the global maximum of $\bar{\rho}_h(r_h)$. Conditions (ii) and (iii) are the boundary conditions ensuring that $\bar{\rho}_h(r_h)$ is not maximized at $r_h^{\min}$ or $r_h^{\max}$, respectively. In particular, a sufficiently large γ ensures the unit return function is not maximized at $r_h^{\max}$, while a sufficiently low density of f_h at $\underline{\theta}$ rules out a maximum at $r_h^{\min}$.

The intuition is as follows. As shown in equation (3), both the payoff from non-defaulted loans and the expected loss from default increase with the loan rate r_h . The latter reflects a negative adverse-selection effect on the average loan return $\bar{\rho}_h$ as higher loan rates induce riskier borrower composition. When γ is large, collateral valuation upon default is costly, so the marginal increase in expected default losses dominates the marginal gain from higher loan repayments. As a result, $\bar{\rho}_h(r_h)$ decreases in r_h at $r_h^{\max}$. On the other hand, a sufficiently small $f_h(\underline{\theta})$ implies that there are few low-risk borrowers.

A marginal increase in the loan rate at $r_h^{\min}$ therefore does not materially deteriorate the risk composition of the loan portfolio, while it raises the return on non-defaulted loans. Hence, the unit average return increases in r_h at $r_h^{\min}$.

This lemma demonstrates the adverse selection effect in [Stiglitz and Weiss \(1981\)](#). This finding contrasts with the conclusion of [Arnold and Riley \(2009\)](#), as we incorporate a collateral valuation cost γ . At the return-maximizing rate $\hat{r}_h$, the supply of bank loans may fall short of the demand from entrepreneurs who are not screened out by the interest rate, which gives rise to credit rationing at the intensive margin.⁴

For type- l entrepreneurs, with probability δ , loan repayment is subsidized by a third party in the event that the R&D project fails. Hence, the probability of failing to repay the loan becomes $\theta(1-\delta)$. The unit average return on LDRF loan is

$$\begin{aligned}\bar{\rho}_l(r_l) &= \frac{\int_{\hat{\theta}_l(r_l)}^{\bar{\theta}} [(1-\theta(1-\delta))r_l + \theta(1-\delta)(b_l\kappa_l - \gamma)] dF_l(\theta)}{1 - F_l(\hat{\theta}_l(r_l))} \\ &= r_l - (1-\delta)\mathbb{E}\left[\theta|\theta \geq \hat{\theta}_l(r_l)\right][r_l - (b_l\kappa_l - \gamma)].\end{aligned}\quad (4)$$

LEMMA 3. Denote $\hat{\delta}_l = 1 - \frac{2}{2\theta + (1-\theta)[1 + (1-\theta)\gamma/(\mu_l - b_l\kappa_l)]}$. If $G_l''(x) \geq \frac{2}{1-x}G_l'(x)$ and $\delta \geq \hat{\delta}_l$, then the unit loan return $\bar{\rho}_l(r_l)$ strictly increases in r_l .

This lemma shows that the credit rationing problem does not arise in the type- l loan market as long as the third party's default subsidy guarantee is sufficiently strong. Banks are willing to lend at the highest possible interest rate, ensuring the market clears. Thus the return-maximizing rate is $\hat{r}_l = r_l^{\max}$.

For our purpose, we denote r_o^m and L_o^m as the revenue-maximizing loan rate and credit supply of a monopolist in the price subgame of the type- o market. That is,

$$r_o^m \in \arg \max_{r \in [0, r_o^{\max}]} \{\bar{\rho}_o(r) \cdot D_o(r)\}, \text{ and } L_o^m = D_o(r_o^m), \quad (5)$$

where $D_o(r_o) = \lambda_o \bar{z}[1 - F_o(\hat{\theta}_o(r_o))]$ is the aggregate loan demand. The following lemma characterizes the uniqueness and property of the revenue-maximizing rate.

⁴ We consider equilibria in which banks grant loans to all borrowers submitting applications. This assumption reflects banks' preference for portfolio diversification and their compliance with regulatory borrower concentration limits. For a discussion, see [Diamond \(1984\)](#) and [Rao et al. \(2015\)](#).

LEMMA 4. Suppose $G_o''(x) \geq \frac{2}{1-x}G_o'(x)$, there is a unique r_o^m such that $r_o^{min} \leq r_o^m \leq \hat{r}_o$. If $\hat{r}_o > r_o^{min}$, then $r_o^m < \hat{r}_o$. If $f_o(\underline{\theta}) < \frac{\mu_o - b_o \kappa_o}{(1-\underline{\theta})^2} \frac{1 - (1-\delta)\mathbb{E}[\theta_o]}{b_o \kappa_o - \gamma + \left(\frac{\mu_o - b_o \kappa_o}{(1-\underline{\theta})} + \gamma\right)(1-\underline{\theta}(1-\delta_o))}$, then $r_o^m > r_o^{min}$.

Lemmas 1 to 4 characterize the sufficient conditions for the properties of loan return and revenue functions used in the equilibrium analysis, and the proofs can be found in Online Appendix B.2. We summarize the assumptions underlying these lemmas below; they will apply throughout the remainder of theoretical analysis.

ASSUMPTION 1. Assume that for $o \in \{h, l\}$, (i) $\mu_o > b_o \kappa_o$; (ii) $G_o''(x) \geq \frac{2}{1-x}G_o'(x)$; (iii) $f_h(\underline{\theta}) \left[1 + \frac{(1-\underline{\theta})\gamma}{\mu_h - b_h \kappa_h}\right] < \frac{1 - \mathbb{E}[\theta_h]}{(1-\underline{\theta})(\mathbb{E}[\theta_h] - \underline{\theta})}$; (iv) $\gamma > \frac{\mu_h - b_h \kappa_h}{1-\underline{\theta}}$; (v) $\delta \geq \hat{\delta}_l$; (vi) $f_o(\underline{\theta}) < \frac{\mu_o - b_o \kappa_o}{(1-\underline{\theta})^2} \frac{1 - (1-\delta)\mathbb{E}[\theta_o]}{b_o \kappa_o - \gamma + \left(\frac{\mu_o - b_o \kappa_o}{(1-\underline{\theta})} + \gamma\right)(1-\underline{\theta}(1-\delta_o))}$.

Although we impose Assumption 1 to facilitate equilibrium characterization, the underlying requirements are more general. Specifically, we require that the HDRF loan return $\bar{\rho}_h(r_h)$ be hump-shaped with a unique interior global maximum; that the LDRF loan return $\bar{\rho}_l(r_l)$ be monotonically increasing when δ is sufficiently large; and that the revenue functions in both markets be hump-shaped with unique interior global maxima. Importantly, these properties do not hinge on Assumption 1 per se. In particular, the condition $G_o''(x) \geq \frac{2}{1-x}G_o'(x)$ serves only as a sufficient condition for the uniqueness results. Accordingly, in the proofs of propositions in Sections 3.3 to 3.5, we provide more general conditions for ensuring that the equilibrium is unique. In Section 5, we further demonstrate numerically that the same qualitative features arise under a broad class of distributions, including tuned beta distributions and truncated normal distributions, for which the condition $G_o''(x) \geq \frac{2}{1-x}G_o'(x)$ need not hold. This indicates that our results are robust to alternative specifications of borrower risk heterogeneity.

3.3. Cournot Competition

To build intuition, we first consider a Cournot benchmark. In this setting, banks simultaneously choose loan capacities, taking as given their competitors' capacities and the inverse loan demand functions, while loan interest rates adjust to clear the market given aggregate capacity. This benchmark can be viewed as a restricted version of the two-stage game in which pricing is competitive in the second stage.

A detailed characterization of the Cournot benchmark is provided in Online Appendix B.4.1. Under Assumption 1, the best-response function is unique and well behaved: it is decreasing in competitors' capacity and in the bank's own capacity in the other market. Denote by L_o^c the symmetric equilibrium

capacity and by $r_o^c = D_o^{-1}(NL_o^c)$ the corresponding equilibrium loan rate, for $o \in \{h, l\}$. We refer to these as the Cournot capacity and Cournot price.

This benchmark is useful for interpreting the full model. In the absence of adverse selection, the Cournot outcome is the unique equilibrium of the two-stage game (Kreps and Scheinkman 1983). In Section 3.4, we show that this result extends to environments with sufficiently weak adverse selection, for example, when the loan default subsidy for LDRFs is sufficiently strong. By contrast, in the HDRF market, equilibrium may instead involve credit rationing.

3.4. Equilibrium Characterization

We solve the equilibrium backward, presenting key results in the main text and relegating the full characterization and proofs to Online Appendix B.4.2. Existence of an equilibrium in the second-stage price subgame follows from Dasgupta and Maskin (1986). The analysis below characterizes how the equilibrium in the price subgame depends on banks' first-stage loan capacities.

We define an equilibrium of the price subgame in type- o market as a *market-clearing equilibrium* if all banks choose the market clearing rate $R_o(\sum_j L_{o,j})$ in pure strategies, where $R_o(\cdot)$ denotes the inverse loan demand function. Let $\hat{L}_o \equiv D_o(\hat{r}_o)$ be the aggregate loan demand at the return-maximizing rate $\hat{r}_o$. We define an equilibrium of the price subgame as a *credit-rationing equilibrium* if all banks choose $\hat{r}_o$ in pure strategies and the aggregate loan capacity satisfies $\sum_j L_{o,j} < \hat{L}_o$.

The following proposition characterizes the conditions for existence and uniqueness of pure-strategy equilibrium in the second stage.

PROPOSITION 1. *Let banks' loan capacities be fixed. (i) The price subgame has a pure-strategy equilibrium if $\sum_j L_{o,j} \leq L_o^m$, where it is a market-clearing equilibrium if $\hat{L}_o \leq \sum_j L_{o,j} \leq L_o^m$, and is a credit-rationing equilibrium if $\sum_j L_{o,j} < \hat{L}_o$; otherwise, it has a mixed-strategy equilibrium if $\sum_j L_{o,j} > L_o^m$. (ii) If a pure-strategy equilibrium exists, it is the unique equilibrium for the price subgame.*

The intuition is as follows. In any pure-strategy equilibrium, all banks must set the same rate. If one bank charged a strictly lower rate than others, it would still lend up to capacity, but could profitably raise its rate slightly while continuing to attract the same borrowers. Conversely, a bank charging a strictly higher rate would face only residual demand and would want to undercut its rivals. Hence unequal rates cannot be sustained in pure strategies.

The key determinant of the second-stage outcome is aggregate loan capacity. When total capacity is low, banks remain capacity constrained even at the return-maximizing rate $\hat{r}_o$. In this case, lowering the loan rate cannot increase lending, since each bank is already constrained by capacity, while

raising the rate reduces unit return. Hence all banks optimally charge $\hat{r}_o$, and equilibrium features credit rationing.

When aggregate capacity is higher, but still below L_o^m , the unique pure-strategy equilibrium is market clearing. At the market-clearing rate $R_o(\sum_j L_{o,j})$, each bank again lends up to its capacity. A unilateral rate cut only lowers return per loan without increasing lending. Under our random-rationing assumption, a unilateral rate increase leaves the deviating bank with only residual demand, which is proportional to aggregate demand D_o . The deviating bank's revenue is therefore proportional to monopoly revenue, which is maximized at $r_o^m < R_o(\sum_j L_{o,j})$ when aggregate capacity is below L_o^m . A unilateral rate increase is therefore also unprofitable. As in the capacity-constrained pricing logic of [Kreps and Scheinkman \(1983\)](#) and [Herk \(1993\)](#), this is precisely the region in which pure-strategy pricing is sustainable.

By contrast, when aggregate capacity exceeds L_o^m , a pure-strategy equilibrium breaks down. In this region, $r_o^m > R_o(\sum_j L_{o,j})$, so a bank can profitably raise its rate above the market-clearing level and still retain positive residual demand. But if banks set rates above the market-clearing level, each then has an incentive to undercut slightly in order to move ahead in borrower allocation and expand lending, potentially up to full capacity. These opposing incentives rule out a pure-strategy outcome and lead instead to mixed-strategy pricing.

In the first stage, each bank chooses its capacities to maximize second-stage equilibrium revenue net of total capacity costs. Let $\pi_{o,i}(L_{o,i}, \mathbf{L}_{o,-i})$ denote bank i 's expected revenue in the price subgame for type- o market, where $\mathbf{L}_{o,-i}$ is the vector of capacities chosen by the other banks. bank i 's problem is

$$\max_{L_{h,i}, L_{l,i}, R_i} \sum_{o \in \{h,l\}} [\pi_{o,i}(L_{o,i}, \mathbf{L}_{o,-i}) - c_o(L_{o,i})] + \rho_r R_i - c_r(R_i) - \bar{r}_d D(N), \quad (6)$$

$$\text{s.t. } L_{h,i} + L_{l,i} + R_i = D(N). \quad (7)$$

Denote bank i 's capacity best-response correspondence as $\tau_o(\mathbf{L}_{h,-i}, \mathbf{L}_{l,-i})$. Following [Herk \(1993\)](#), [Acemoglu et al. \(2009\)](#) and others, we focus on pure-strategy equilibrium in the first stage. A subgame-perfect equilibrium of the capacity game is therefore a collection of capacities $\{L_{o,i}^*\}_{i=1}^N$ for which $L_{o,i}^* \in \tau_o(\mathbf{L}_{h,-i}^*, \mathbf{L}_{l,-i}^*) \forall i$.

We further restrict attention to equilibrium paths for which the second-stage price subgame admits a pure-strategy equilibrium, thereby ruling out mixed-strategy pricing. This restriction is satisfied when the number of banks, N , is sufficiently small. The intuition is that, as in the Cournot benchmark, aggregate loan capacity tends to increase with the number of banks. According to [Proposition 1](#), once aggregate capacity becomes too large, price competition becomes too intense to sustain a pure-strategy outcome,

and mixed-strategy pricing arises instead. The following lemma provides a sufficient condition under which aggregate loan capacity greater than L_o^m cannot constitute a subgame-perfect equilibrium.

LEMMA 5. *Suppose*

$$\bar{\rho}_o(r_o^m) < \frac{2}{L_o^m} [c_o(L_o^m) + c_r(\bar{D} - L_o^m) - c_r(\bar{D})] + 2\rho_r. \quad (8)$$

Let $\bar{N}_o$ denote the smallest solution for $N \geq 1$ to the equation

$$\bar{\rho}_o(r_o^m) = \frac{N+1}{L_o^m} \left[c_o\left(\frac{L_o^m}{N}\right) + c_r\left(D(N) - \frac{L_o^m}{N}\right) - c_r(D(N)) \right] + \frac{N+1}{N} \rho_r. \quad (9)$$

If no such solution exists, define $\bar{N}_o = +\infty$. For any $N < \bar{N}_o$, an aggregate loan capacity $\sum_i L_{o,i} \geq L_o^m$ cannot constitute a subgame-perfect equilibrium in type-o market.

The following proposition characterizes the conditions for the uniqueness of subgame-perfect equilibrium in which the unit loan return for HDRFs is hump-shaped and the unit loan return for LDRFs is monotonically increasing. This unique equilibrium is symmetric where banks build the same loan capacities and post the same loan contracts.

PROPOSITION 2. *Suppose Assumption 1 holds, $N < \min\{\bar{N}_h, \bar{N}_l\}$, and for each $o \in \{h, l\}$, assume $\rho_r + c'_o(0) < \bar{\rho}_o(\hat{r}_o)$,*

$$\bar{\rho}_o(r_o^m) < \frac{2}{L_o^m} [c_o(L_o^m) + c_r(\bar{D} - L_o^m) - c_r(\bar{D})] + 2\rho_r. \quad (10)$$

Denote function $F_l(L_l, L_h, N)$, $F_h(L_l, L_h, N)$ as

$$F_l(L_l, L_h, N) = \bar{\rho}'_l(\min\{\hat{r}_l, R_l(NL_l)\})R'_l(NL_l)L_l + \bar{\rho}_l(\min\{\hat{r}_l, R_l(NL_l)\}) - c'_l(L_l) - \rho_r + c'_r(D(N) - L_l - L_h), \quad (11)$$

$$F_h(L_l, L_h, N) = \bar{\rho}'_h(\min\{\hat{r}_h, R_h(NL_h)\})R'_h(NL_h)L_h + \bar{\rho}_h(\min\{\hat{r}_h, R_h(NL_h)\}) - c'_h(L_h) - \rho_r + c'_r(D(N) - L_l - L_h). \quad (12)$$

Let $\tilde{L}_l(N)$ be the solution for L_l to $F_h(L_l, \hat{L}_h/N, N) = 0$. If $\lim_{L \rightarrow +\infty} F_h(L, \hat{L}_h/N, N) \geq 0$, set $\tilde{L}_l(N) = +\infty$; if $\lim_{L \rightarrow -\infty} F_h(L, \hat{L}_h/N, N) \leq 0$, set $\tilde{L}_l(N) = -\infty$.

There exists a unique subgame-perfect equilibrium such that all banks choose the same loan capacity L_l^*, L_h^* given by

$$F_l(L_l^*, L_h^*, N) \leq 0, \text{ with equality if } L_l^* > 0, \quad (13)$$

$$F_h(L_l^*, L_h^*, N) \leq 0, \text{ with equality if } L_h^* > 0. \quad (14)$$

Moreover, the equilibrium has the following property:

- (i) If $F_h(0, \hat{L}_h/N, N) \geq 0$ and $F_l(\tilde{L}_l(N), \hat{L}_h/N, N) \leq 0$, both the LDRF and HDRF markets exhibit a market-clearing equilibrium. Banks choose Cournot capacity $(L_l^*, L_h^*) = (L_l^c, L_h^c)$, set the Cournot prices (r_l^c, r_h^c) and common credit limits $(\bar{z}_l, \bar{z}_h)$. The loan capacity satisfy $L_l^* < L_l^m/N$ and $L_h^* \in [\hat{L}_h/N, L_h^m/N)$.
- (ii) Otherwise, the HDRF exhibits a credit-rationing equilibrium and the LDRF market exhibits a market-clearing equilibrium. Banks set loan rates $(R_l(NL_l^*), \hat{r}_h)$, loan capacities satisfy $L_l^* < L_l^m/N$, $L_h^* < \hat{L}_h/N$, and credit limits $(\bar{z}_l, \hat{z}_h^*)$ satisfying

$$NL_h^* = \lambda_h \left[1 - F_h \left(\hat{\theta}_h(\hat{r}_h) \right) \right] \hat{z}_h^*. \quad (15)$$

- (iii) If the cost functions are quadratic, i.e., $c_h(L_h) = \frac{\chi_h}{2} L_h^2$, $c_l(L_l) = \frac{\chi_l}{2} L_l^2$, and $c_r(R) = \frac{\chi_r}{2} R^2$, and if $(\chi_h + \chi_r)\hat{L}_h \geq \chi_r \bar{D}$, there exists a unique threshold N_h^* such that the HDRF market features a market-clearing (credit-rationing) equilibrium if and only if $N \geq (<) N_h^*$.

In this proposition, conditions (13) and (14) are the first-order conditions for individual bank profit maximization in equilibrium. For type- o loans, the marginal benefits of loan issuance consist of marginal revenue, $\bar{\rho}'_o R'_o L_o + \bar{\rho}_o$, and the marginal reduction in reserve management costs, $c'_r(D(N) - L_l^* - L_h^*)$. The marginal costs include the marginal loan capacity cost, c'_o , and the funding cost ρ_r . Banks equate these marginal benefits and costs when choosing loan supply.

It is worth noting that the reserve-cost-saving term c'_r captures two additional forces arising from multi-market competition relative to a single-market framework. First, the term c'_r depends on both L_l^* and L_h^* , reflecting a substitution effect across loan markets: expanding lending in one market reduces the marginal benefit of lending in the other. Second, c'_r also depends on deposits $D(N)$. Greater deposit competition lowers individual deposits and thus reduces the marginal cost-saving from reserve management, which in turn dampens loan issuance. Due to these additional forces, the effects of greater bank competition on loan allocation may be non-monotonic, as shown in Section 3.5.

Regarding the equilibrium firm borrowing, Proposition 2 shows that the LDRF borrowers are fully financed, and the threshold borrower type, $\hat{\theta}_l$, which corresponds to the extensive margin of innovation investment, is a function of bank competition N . In the HDRF market, by contrast, entrepreneurs are not fully financed under low N due to credit rationing, resulting in an equilibrium loan rate that is independent of bank competition. Consequently, the market-clearing condition in the HDRF market determines the individual loan size, corresponding to the intensive margin of innovation investment.

However, this credit-rationing equilibrium persists only if the number of banks is small enough to satisfy $N < N_h^*$. When N becomes large, aggregate

loan capacity eventually exceeds loan demand at $\hat{r}_h$, eliminating rationing. In this case, intensified bank competition in HDRF market induces a shift to a market-clearing equilibrium with Cournot pricing.

3.5. Effects of Banking Competition

We now derive the main comparative statics.⁵ To make the mechanism transparent, we begin with the empirically relevant case in which the LDRF market clears while the HDRF market is credit-rationed. In that region, competition affects LDRFs primarily through loan rates and participation, but affects HDRFs primarily through loan size and investment scale. This is the key reason the model maps naturally into extensive- and intensive-margin outcomes in the data.

3.5.1. Loan Resource Allocation

According to Proposition 2, for LDRFs the equilibrium loan rate r_l^* depends on the number of banks N . Bank competition therefore affects LDRF loan allocation through the extensive margin: changes in r_l^* shift the marginal riskiness type $\hat{\theta}_l$ and hence the fraction of loan market participants $1 - F_l(\hat{\theta}_l(r_l^*))$. By contrast, the individual loan size is fixed at $\bar{z}_l$, so the intensive margin of LDRFs remains constant and independent of N .

For HDRFs, equilibrium condition (15) implies that aggregate loan supply varies with N , while the loan rate remains fixed at $\hat{r}_h$. Thus, bank competition affects HDRF loan allocation along the intensive margin through the individual loan size $\hat{z}_h^*$. Meanwhile, the extensive margin remains constant at $\hat{\theta}_h(\hat{r}_h)$, independent of N .

Consistent with the standard Cournot intuition that greater competition lowers prices and expands aggregate supply, an increase in N should raise the extensive margin for LDRFs via a lower loan rate, and increase the intensive margin for HDRFs via higher loan size. The following proposition shows that this result holds when the curvature of reserve management costs $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small.

PROPOSITION 3. *If $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small, a higher degree of bank competition increases the extensive margin of LDRF borrowing and R&D investment, and increases the intensive margin of HDRF borrowing and R&D investment. Consequently, the aggregate loan resources for R&D financing increase.*

The intuition behind the proposition is straightforward. In the LDRF loan market, equilibrium is market-clearing, with banks competing à la Cournot in capacity. As long as a project is profitable, each entrepreneur demands a

⁵ Proofs of propositions in this section can be found in Online Appendix B.4.2.

fixed loan size $\bar{z}_l$. Greater bank competition, via an increase in the number of banks, leads to a lower equilibrium loan rate. This expands the set of LDRFs with profitable projects, thereby increasing the extensive margin of R&D investment.

In the HDRF loan market, by contrast, the loan rate is fixed at $\hat{r}_h$ due to credit rationing. Here, aggregate loan supply depends on the number of banks. A larger number of banks increases total loan capacity, raising the intensive margin of HDRF borrowing and R&D investment, provided that the unit return on HDRF loans remains above the marginal capacity cost.

Note that Proposition 3 does not necessarily hold when $c_r''(D(N) - L_h^* - L_l^*)$ is large. The intuition is as follows. A key marginal benefit of loan issuance is the reduction in reserve management cost, captured by $c_r'(D(N) - L_l^* - L_h^*)$. An increase in N reduces individual deposits $D(N)$, which in turn reduces this marginal cost saving. When the curvature $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently large, the decline in $c_r'(D(N) - L_l^* - L_h^*)$ dominates, thereby reducing the overall marginal benefit of loan issuance and lowering individual bank's loan supply. Consequently, both the extensive margin of LDRF borrowing and the intensive margin of HDRF borrowing may decrease with N in this case. Moreover, a large $c_r''(D(N) - L_h^* - L_l^*)$ implies a strong substitution effect between the two types of lending. In this case, an increase in N may raise aggregate loan supply in one market while reduce it in the other. In Online Appendix B.4.2, the proof of Proposition 3 provides detailed conditions (B28) and (B29) under which aggregate loan supply in each market increases with N .

Next, we examine how bank competition affects loan allocation between the two firm types. The loan allocation ratio is defined as L_l^*/L_h^* . The following proposition shows that the loan allocation ratio decreases as bank competition intensifies.

PROPOSITION 4. *If $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small, we have $\frac{d}{dN} \left(\frac{L_l^*}{L_h^*} \right) < 0$, indicating that loan resources shift increasingly toward HDRFs as bank competition intensifies.*

Intuitively, from (13) and (14), when $c_r''(D(N) - L_h^* - L_l^*)$ is small, the equilibrium loan capacities are driven primarily by marginal loan revenues. For HDRF loans, the marginal revenue is $\bar{\rho}_h(\hat{r}_h)$, which is independent of N . In contrast, the marginal revenue for LDRF loans is $\bar{\rho}_l'(R_l(NL_l))R_l'(NL_l)L_l + \bar{\rho}_l(R_l(NL_l))$, which decreases with N . This implies that in HDRF loan capacity L_h^* is relatively insensitive to changes in N , whereas LDRF loan capacity L_l^* decreases significantly with N . Therefore, the loan allocation ratio declines with N in this case.

3.5.2. Risk-taking

We analyze the effects of bank competition on risk-taking by considering the non-performing loan (NPL) share, defined as the proportion of defaulted loans in total loans. For LDRFs and HDRFs, the NPL shares are given by

$$\text{NPL}_l = (1 - \delta) \mathbb{E}[\theta_l \mid \theta_l \geq \hat{\theta}_l(r_l^*)], \quad \text{NPL}_h = \mathbb{E}[\theta_h \mid \theta_h \geq \hat{\theta}_h(r_h^*)], \quad (16)$$

where δ is the probability of government subsidy upon default, which lowers the NPL share for LDRFs. The expectations $\mathbb{E}[\theta_l \mid \theta_l \geq \hat{\theta}_l(r_l^*)]$ and $\mathbb{E}[\theta_h \mid \theta_h \geq \hat{\theta}_h(r_h^*)]$ represent the conditional default probability based on the equilibrium loan rates r_l^* and r_h^* . Given equilibrium loan allocations L_l^* and L_h^* , the aggregate NPL share is

$$\text{NPL} = \frac{L_l^*}{L_l^* + L_h^*} (1 - \delta) \mathbb{E}[\theta_l \mid \theta_l \geq \hat{\theta}_l(r_l^*)] + \frac{L_h^*}{L_l^* + L_h^*} \mathbb{E}[\theta_h \mid \theta_h \geq \hat{\theta}_h(r_h^*)]. \quad (17)$$

PROPOSITION 5. *A higher degree of bank competition has no effect on the NPL share for HDRFs. By contrast, it reduces the NPL share for LDRFs when $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small.*

This proposition highlights the heterogeneous effects of bank competition on bank risk-taking in lending to LDRFs and HDRFs. The key mechanism operates through the extensive margin, because the average default probability is pinned down by the riskiness of the marginal borrower. Since only the extensive margin for LDRFs moves with N in this regime, greater bank competition reduces risk-taking in LDRF loans, while the risk in HDRF loans remains unchanged. However, loan allocation shifts toward HDRFs, whose loans are inherently riskier due to the absence of government default subsidies, so the net effect on the aggregate NPL share is ambiguous. This is precisely the distinction we want to emphasize relative to the existing literature: aggregate portfolio risk can rise even though screening does not worsen and within-segment loan quality does not deteriorate.

3.5.3. R&D Output and Profit Allocation

The aggregate net R&D output is given by:

$$\lambda_l \left[1 - F_l \left(\hat{\theta}_l(r_l^*) \right) \right] (\mu_l - 1) z_l^* + \lambda_h \left[1 - F_h \left(\hat{\theta}_h(r_h^*) \right) \right] (\mu_h - 1) z_h^*. \quad (18)$$

From Proposition 3, since $\hat{\theta}_l(r_l^*)$ decreases and z_h^* increases with N , while z_l^* and $\hat{\theta}_h(r_h^*)$ unchanged, we obtain the following result:

PROPOSITION 6. *When $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small, R&D output increases with the degree of bank competition.*

Intuitively, greater bank competition expands the extensive margin of LDRF borrowing and the intensive margin of HDRF borrowing, thereby increasing total R&D investment. As a result, the R&D output rises monotonically with the degree of bank competition when the curvature $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small.

We next examine profit allocation. The aggregate R&D profits of LDRFs and HDRFs are:

$$\Pi_l = \lambda_l \int_{\hat{\theta}_l(r_l^*)}^{\bar{\theta}} v_l(\theta, r_l^*) f_l(\theta) d\theta \cdot z_l^*, \quad (19)$$

$$\Pi_h = \lambda_h \int_{\hat{\theta}_h(r_h^*)}^{\bar{\theta}} v_h(\theta, r_h^*) f_h(\theta) d\theta \cdot z_h^*. \quad (20)$$

Consider the case where $c_r''(D(N) - L_h^* - L_l^*)$ is sufficiently small. Since $v_l(\theta, r_l^*)$ decreases in r_l^* , r_l^* falls with N , and z_h^* increases with N , both Π_l and Π_h increase in N .

The aggregate bank profits are defined as:

$$\begin{aligned} \Pi_b = N [\bar{\rho}_l(r_l^*)L_l^* + \bar{\rho}_h(r_h^*)L_h^* - c_l(L_l^*) - c_h(L_h^*) \\ + \rho_r(D(N) - L_l^* - L_h^*) - c_r(D(N) - L_l^* - L_h^*) - \bar{r}_d D(N)]. \end{aligned} \quad (21)$$

Using banks' first order condition of optimal loan capacity, we derive:

$$\frac{d\Pi_b}{dN} = \frac{1}{N} \Pi_b + N L_l^* \bar{\rho}_l' R_l' \frac{d[(N-1)L_l^*]}{dN} + N(\rho_r - c_r' - \bar{r}_d) D'(N). \quad (22)$$

The first term is positive and reflects the scale effect: higher competition expands lending and raises profits through greater volume. The second term is negative and captures profit margin erosion from intensified competition in the LDRF market, which operates through a lower loan rate. By contrast, the profit margin on HDRF loans does not vary with N because the equilibrium loan rate is fixed. The third term represents the marginal effect of deposit competition. Banks earn the spread between the return on reserves and the marginal cost of deposit issuance; when this spread is positive, greater deposit competition reduces profits, making this term negative. Intuitively, stronger bank competition increases loan supply to both LDRFs and HDRFs, thereby raising entrepreneurs' profits through higher R&D investment. For banks, however, the net effect reflects a trade-off between a positive scale effect and a negative competition effect. Whether total bank profits rise or fall with competition depends on which force dominates.

4. Empirical Study

This section takes the model's central predictions to the data. The empirical objective is not simply to ask whether banking competition raises average innovation, but to test whether it does so through different margins for

different borrower types. Guided by Proposition 3, we examine whether stronger bank competition increases the probability of R&D investment for the relatively lower-risk group and the scale of R&D investment conditional on participation for the relatively higher-risk group. In the Chinese setting, ownership provides a natural empirical proxy for that segmentation: SOFs are treated as low-default-risk firms because of implicit support, whereas PFs are treated as high-default-risk firms because they face tighter credit constraints and weaker protection.

4.1. *Institutional Background*

This subsection summarizes the institutional features needed to interpret the empirical design. Two features matter most. First, bank discrimination and relationship lending were pervasive, so private firms (PFs) often faced tighter credit conditions than state-owned firms (SOFs) (Cull and Xu 2003, Ge and Qiu 2007, Guariglia et al. 2011). Second, SOFs could benefit from explicit or implicit government support in distress, whereas PFs generally could not. These features motivate our empirical mapping from ownership to default risk and make China a natural setting for studying how banking competition interacts with borrower heterogeneity. We then place the 2006 deregulation episode in the longer evolution of China's banking system by distinguishing three broad phases of commercial-bank development.⁶

The first stage, spanning from 1949 to 1978, saw the monopolization of the financial sector by the People's Bank of China (PBOC), which was owned and controlled by the Ministry of Finance. During this period, there was virtually no competition among banks. The PBOC controlled approximately 93% of the country's total financial assets and managed nearly all financial transactions (Allen et al. 2008). This situation changed significantly after 1978, when reforms altered the banking structure.

The second stage of China's banking development, spanning from 1978 to 1995, marked a significant enhancement in banking competition. During this period, the Chinese government moderately enhanced banking competition by establishing five major state-owned commercial banks and introducing various financial intermediaries into the credit market. These intermediaries comprised policy banks, rural and urban credit cooperatives, joint-stock banks, and non-bank financial institutions. The four largest state-owned banks included the Bank of China (BOC), the People's Construction Bank of China (PCBC), the Agricultural Bank of China (ABC), and the Industrial and Commercial Bank of China (ICBC), which was established in 1984 and assumed responsibility for all commercial transactions previously managed

6 Lin et al. (2015) also classified the development of China's banking sector after 1949 into three stages. However, our focus here is on the discussion of China's commercial banks, as they are the primary providers of loans to firms. For a comprehensive analysis of China's financial development prior to 1949, see Allen et al. (2008).

by the People's Bank of China (PBOC). Since their establishment, these four banks have consistently been the largest in China's banking system, alongside the Bank of Communications (BOCOM), established in 1986 as a state-owned joint-stock commercial bank. In addition to the five major state-owned banks, several joint-stock commercial banks emerged during this period, further intensifying competition within the banking sector. Regional banks were also created in Special Economic Zones along the coast, and both Rural Credit Cooperatives (RCCs) and Urban Credit Cooperatives (UCCs) were established (Allen et al. 2008). The implementation of the Commercial Bank Law of China in 1995 marked the conclusion of this second stage, redefining state-owned banks as "commercial banks" and requiring them to operate based on market principles rather than relying solely on policy lending. Overall, the second stage solidified the presence of five major commercial banks and numerous smaller joint-stock banks in China's banking sector.

The third stage of China's banking development was characterized by significant deregulation efforts. A defining feature of this stage was the transformation of Urban Credit Cooperatives (UCCs) into city commercial banks, a shift that significantly increased competition within the banking sector. In 1995, the establishment of the first city commercial bank, Shenzhen City Cooperation Bank, marked a pivotal step in serving small and medium-sized enterprises and local residents. By 2017, the number of city commercial banks had increased to 134. A key development during this period was the establishment of the China Banking Regulatory Commission (CBRC), which has since initiated a series of reforms aimed at eliminating restrictive banking regulations. In 2006, the CBRC issued a policy document titled *Management Methods on City Commercial Bank Branches Outside the Headquarters*, facilitating city commercial banks' operations across multiple cities under the oversight of the local CBRC office. In 2009, the CBRC further relaxed restrictions, allowing both joint-stock banks and city commercial banks to establish branches nationwide. This development created exogenous variations in local banking competition, thereby facilitating an evaluation of its effects on innovation.

4.2. Data

4.2.1. Data Source

We use two datasets. The first is the Chinese Manufacturing Firms Database (CMFD), compiled by the National Bureau of Statistics, which covers 2006–2007 and provides firm-level information on ownership, location, financial characteristics, and R&D investment.

Our second dataset is a city-level census of bank branches collected by the China Banking Regulatory Commission (CBRC) for 2006–2011. Because the firm-level data are available only for 2006 and 2007, we restrict the merged sample to those two years. The branch dataset covers 336 cities in 31

provincial-level regions and classifies financial institutions into 14 categories. We focus on commercial banks, which are the main providers of credit to firms and therefore the most relevant institutions for our analysis of innovation finance. These banks account for approximately 44% of total loans and 41% of total employment among all financial institutions in the dataset. A limitation of this dataset is that the information is relatively aggregated. Our final sample comprises 361,394 observations, including 161,869 firms in 2006 and 199,525 firms in 2007.

4.2.2. Key Variables and Correlations

We next define the three core variables used in the empirical analysis: banking competition, innovation investment, and firm ownership.

Banking Competition: Herfindahl-Hirschman Index (HHI) and CR5

Bank competition is measured using the market shares of banks in the local credit market. China's banking system comprises state-owned, joint-stock, and city commercial banks. Following the methodologies of Degryse and Ongena (2007) and Chong et al. (2013), we construct a city-level index of bank competition. We evaluate the intensity of bank competition using the Herfindahl-Hirschman Index (HHI), treating the share of bank loans within a city as the market share for each bank. The HHI index for city c in year t is constructed as follows:

$$HHI_{ct} = \sum_{n=1}^3 \left(\frac{loan_{cnt}}{\sum_{n=1}^3 loan_{cnt}} \right)^2 \quad (23)$$

In this equation, n denotes the three types of banks in the city: $n = 1$ for the five largest state-owned banks, $n = 2$ for joint-stock banks, and $n = 3$ for city commercial banks. Here, $loan_{cnt}$ represents the total amount of loans issued by banks of type n in city c during year t . The index HHI_{ct} ranges from 0 to 1, equating to 1 in a monopolistic loan market and approaching 0 in a perfectly competitive market.

Table 1 presents the summary statistics for HHI_{ct} in our sample. The mean HHI_{ct} remains relatively stable throughout the sample period, while the minimum competition index declines from 0.408 to 0.398, and the median decreases slightly from 0.698 to 0.696.

China's banking sector is dominated by the five largest state-owned banks. These banks dominate the market for deposits and loans. Therefore, the decrease in the market power of these banks indicates a more competitive loan market. Consequently, we employ the share of total loans offered by the five largest state-owned banks as a measure of the competitive landscape in China's banking sector. We define the market power index as:

$$CR5_{ct} = \frac{loan_{ct}^{big5}}{loan_{ct}^{total}}, \quad (24)$$

TABLE 1
Summary Statistics of Banks' HHI and CR5

Variables	HHI_{ct}				$CR5$			
	Mean	Min	Max	Median	Mean	Min	Max	Median
2006	.703	.408	1	.698	.799	.496	1	.820
2007	.702	.398	1	.696	.800	.430	1	.814

where $loan_{ct}^{big5}$ represents the total loans provided by the five largest state-owned banks, and $loan_{ct}^{total}$ represents the total loans issued by all banks in the same city. We present the summary statistics of $CR5_{ct}$ in the last four columns of Table 1. The summary statistics exhibit a pattern similar to that of HHI_{ct} .

The banking reform initiated by the China Banking Regulatory Commission (CBRC) in 2006 is expected to significantly impact the level of competition in the banking sector. As our empirical strategy relies on the exogenous changes in banking competition, it is essential to verify that competition did indeed change from 2006 to 2007. To effectively capture the changes in banking competition intensity across cities in China, we present the variations in the Herfindahl-Hirschman Index (HHI) and the Concentration Ratio of the top five banks (CR5) from 2006 to 2007 in Table A1 of Online Appendix A.1. Of the cities analyzed, 145 experienced decreases in the HHI index and 146 experienced decreases in the CR5 index, while 68 cities reported increases in HHI and 67 cities reported increases in CR5. This pattern suggests a reduction in the market power of the five largest state-owned banks, likely due to the deregulation of city commercial banks. Additionally, we illustrate the changes in the HHI in Figure A1 and the CR5 in Figure A2 in Online Appendix A.1.

Measuring Margins of Innovation

We measure a firm's innovation activities through R&D expenditures recorded in the firm-level dataset. We use rd_{it} to represent the observed expenditures on R&D. To assess changes in innovation investment along the extensive margin, we create a dummy variable rdd_{it} , which equals one if the firm invests positively in R&D and zero otherwise. For the intensive margin of R&D investment, we use the variable rd_{it} by focusing on firms with positive R&D investment. Simply using rd_{it} for the entire sample to evaluate changes in response to various factors may be misleading because it contains information

from both the extensive and intensive margins.⁷ In the empirical model, for the intensive margin of R&D investment, for firms with positive R&D expenditures, we normalize their R&D expenditures by the total sales to control for firm size. Specifically, we define the intensive margin of R&D investment as $rdint_{it} = rd_{it}/sales_{it}$, where $sales_{it}$ represents the total sales of firm i in year t .⁸ In our dataset, the mean value of rdd_{it} is approximately 10%, indicating that only 10% of firms engage in innovation investment. Moreover, this value is 10.4% in 2006 and 10.5% in 2007, suggesting a 0.1% increase in the share of innovative firms. In the full sample, the average R&D expenditures exhibit an upward trend, rising from 374.52 thousand Chinese yuan in 2006 to 421.80 thousand Chinese yuan in 2007, an increase of 47.28 thousand Chinese yuan.

Classification of Ownership Structures

We classify firms' ownership types based on the origins of pay-in capital at registration. The six categories of investors include: state, collective investors, legal entities, individuals, foreign investors (excluding those from Hong Kong, Macao, and Taiwan), and investors from Hong Kong, Macao, and Taiwan. Guariglia et al. (2011) classifies firms into three ownership categories: state-owned firms (including those owned by the state and collective investors), foreign-owned firms (from Hong Kong, Macao, Taiwan, and other regions), and private firms (owned by legal entities and individuals). They classify firms based on majority ownership shares; a firm is classified as state-owned if its paid-in capital is at least 50% owned by the state and collective investors. We adopt a classification approach similar to that in Guariglia et al. (2011), focusing on private firms (PFs) and state-owned firms (SOFs) because we believe that the innovation activities of foreign firms are influenced more by their headquarters than by the local credit market. Specifically, we classify a firm as state-owned (or private) if over 95% of its paid-in capital is owned by the state and collective investors (or by legal entities and individuals).

⁷ This can be illustrated as follows, note that the average R&D investment can be decomposed as:

$$\frac{1}{NT} \sum_{i=1}^N \sum_{t=1}^T rd_{it} = \underbrace{\frac{\sum_i \sum_t rdd_{it}}{NT}}_{\text{extensive margin}} \times \underbrace{\frac{\sum_i \sum_t rd_{it} \cdot rdd_{it}}{\sum_i \sum_t rdd_{it}}}_{\text{intensive margin}},$$

where $\frac{\sum_i \sum_t rdd_{it}}{NT}$ represents the probability of innovation (extensive margin) and $\frac{\sum_i \sum_t (rd_{it} \cdot rdd_{it})}{\sum_i \sum_t rdd_{it}}$ is the average R&D spending conditional on firms having positive R&D investment (intensive margin).

⁸ As a robustness check, we also normalize the R&D expenditures using total debt, partly controlling the loan size.

Ultimately, we obtain 14,583 observations for SOFs and 324,979 for PFs. This corresponds to 9,936 SOFs and 195,480 PFs.⁹

Table 2 presents summary statistics for the extensive and intensive margins of R&D investment categorized by ownership type. The results indicate that state-owned firms (SOFs) are more active in innovation than private firms (PFs) along the extensive margin. However, the gap between the two ownership types narrowed in 2007. In contrast, private firms (PFs) invested more in R&D along the intensive margin. These results suggest that SOFs and PFs exhibit differing behaviors regarding the extensive and intensive margins of R&D investment.

4.2.3. Correlation between Banking Competition and Innovation Investment by Ownership

We examine how ownership influences the relationship between banking competition and a firm's participation in innovation activities. Table 2 presents summary statistics for the extensive and intensive margins of R&D spending for state-owned firms (SOFs) and private firms (PFs). In 2006, 18.2% of SOFs had positive R&D investments, compared to just 9.3% of PFs. This gap narrowed in 2007. Furthermore, our findings indicate that SOFs invest more in R&D at the intensive margin. These statistics suggest that, on average, SOFs are more active in R&D investment than PFs.

We also present a series of graphical evidence that illustrate the relationship between banking competition and various margins of R&D investment. Building on our theoretical framework, we further examine how firm ownership influences this relationship. The simple correlations between banking competition and the different margins of R&D investment across ownership structures are generally consistent with our theoretical predictions. All results are provided in Online Appendix A.2 and A.3.

4.3. Empirical Evidence

In this section, we present results on empirical tests on banking competition and margins of R&D investment illustrated in Proposition 3.

4.3.1. Empirical Strategy and Results

Extensive Margin: Probability of Investing in R&D

We begin by investigating the relationship between banking competition and the probability of investing in R&D through the estimation of the following probabilistic model:

$$\mathbf{P}(rdd_{ict} = 1 | \mathbf{X}_{ict}) = \Phi(\beta_0 + \beta_1 HHI_{ct} + \beta_2 PF_{ict} + \boldsymbol{\theta}_0 \mathbf{Z}'_{ict}) \quad (25)$$

9 Our classification strategy for ownership structure influences the final sample size. We also tested other classification thresholds: 50%, 70%, and 90%; the results remain stable across these classifications.

TABLE 2
Margins of R&D Investment by Ownership

Margins	Year	Statistic	Extensive Margin rdd_{it}		Intensive Margin ($rdd_{it} = 1$) rd_{it}			
			SOFs	PFs	SOFs	PFs	SOFs	PFs
2006	mean		0.182	0.093	11,379.95	2,337.42	0.0148	0.0151
	sd.		0.386	0.291	53,260.58	22,784.32	0.0407	0.0376
	obs.		6,363	145,120	1,158	13,518	1,158	13,518
2007	mean		0.165	0.095	14,523.68	2,575.99	0.0154	0.0155
	sd.		0.371	0.294	90,458.66	23,823.78	0.0405	0.0352
	obs.		8,220	179,859	1,356	17,147	1,356	17,147
Total	mean		0.172	0.094	13,075.61	2,470.82	0.0151	0.0153
	sd.		0.378	0.292	75,634.66	23,371.18	0.0406	0.0363
	obs.		14,583	324,979	2,514	30,665	2,514	30,665

NOTES: Extensive margin of R&D investment is computed using all observations, while the intensive margin is based only on observations with positive R&D spending. rd_{it} is the level of R&D expenditures measured in 1,000 Chinese yuan; $rdint_{it}$ is the level of R&D expenditures normalized by sales.

where rdd_{ict} is a dummy variable that equals one when firm i , located in city c , has a positive R&D investment in year t , and zero otherwise. $\Phi(\cdot)$ represents the cumulative distribution function of the standard normal distribution.¹⁰ HHI_{ct} is the city-year level bank competition measured by the HHI index calculated for banks in city c at year t , and PF_{ict} is a dummy variable that equals one when the firm is a private firm and zero otherwise. We include the indicator for private firms in our basic econometric specification, as a substantial body of literature documents differences in organizational efficiency between state-owned firms (SOFs) and non-state-owned firms (non-SOFs) in China (Boardman and Vining 1989, Song et al. 2011). $\mathbf{Z}'_{ict}$ denotes a vector of firm-level and city-level control variables.

To examine the different effects of banking competition on the innovation participation between SOFs and PFs, we specify the following probabilistic model:

$$\begin{aligned} \mathbf{P}(rdd_{ict} = 1 | \mathbf{X}_{ict}) = & \Phi\left(\delta_0 + \delta_1 HHI_{ct} \times PF_{ict} + \delta_2 HHI_{ct} \times SOF_{ict} \right. \\ & \left. + \delta_3 PF_{ict} + \boldsymbol{\theta}_1 \mathbf{Z}'_{ict}\right), \end{aligned} \quad (26)$$

10 Our results remain robust when we employ a logit model, assuming that the error term follows a logistic distribution.

where the interaction terms $\delta_1 HHI_{ct} \times PF_{ict}$ and $\delta_2 HHI_{ct} \times SOF_{ict}$ capture the innovation effect of banking competition on SOFs and PFs, respectively. According to Proposition 3, we expect that the coefficient δ_2 is positive and statistically significant, while the coefficient δ_1 is not statistically significant. We use a set of control variables to control for the firm-level and city-level characteristics that are similar to those in the previous model specification (25).

The baseline results for the extensive-margin R&D investment in response to intensified banking competition are reported in Table 3. As we can see in column (1), the coefficient of HHI_{ct} is negative but not statistically significant. Column (3) shows that the coefficient of HHI_{ct} becomes significantly negative after including other firm-level and city-level controls. This may suggest that the overall impact of banking competition on the probability of R&D investment is positive. As shown in Column (2) and Column (4), whether including additional firm-level and city-level controls or not, only the coefficient of $HHI \times SOF$ is statistically significant at the 1% level. This implies that the intensified banking competition has only induced more SOFs to undertake R&D investment. Columns (3) and (4) show that our estimation results remain stable. Comparing the results in Column (4) with Column (2), we notice that the absolute value of the coefficient of $HHI \times SOF$ decreases from -0.479 to -0.410, indicating that other controls slightly contribute to the effect of banking competition on the participation. Also, note that in all columns, the coefficient of PF is negative and significant, which implies that a larger ratio of SOFs invest in R&D than that of PFs, *ceteris parabus*. This result suggests that SOFs might have better access to the credit market, being more capable of obtaining external finance to invest in innovation projects.

Because our sample is a short panel, we do not observe firms' R&D spending before 2006 or after 2007. Unlike capital investment, which is often lumpy (e.g., Cooper et al. 1999; Winberry 2021), R&D investment tends to be smoother over time because a substantial share consists of wages and salaries for scientists and engineers (Hall and Lerner 2010). According to the 2008 innovation survey conducted by the Ministry of Science and Technology, labor costs account for about 33.9% of total R&D spending. This suggests that R&D investment is less likely to be lumpy than conventional capital investment. If R&D investment were in fact lumpy, however, a short panel could still bias our estimate of the extensive margin.¹¹

11 For instance, a firm may have made a substantial investment in 2005 intended to also cover expenses in 2006. However, if the same firm makes an additional investment in 2007, this could be mis-attributed to the policy stimulus, despite the firm's consistent history of R&D investment. Situations like this can lead to an overestimation of the extensive margin. Similar considerations apply to subsequent years after 2007. While the policy might incentivize certain firms to begin investing in R&D, they may initiate their investments for various reasons after 2007. Such cases would not be reflected in the empirical findings of this paper, leading to an underestimation of the extensive

TABLE 3
 Probit estimates of the extensive margin of R&D investments

Dependent variable:	<i>rdd</i>	<i>rdd</i>	<i>rdd</i>	<i>rdd</i>
HHI	-0.117 (0.0813)		-0.175* (0.0962)	
HHI×SOF		-0.479*** (0.113)		-0.410*** (0.150)
HHI×PF		-0.0919 (0.0817)		-0.160 (0.0980)
PF	-0.239*** (0.0500)	-0.517*** (0.105)	-0.284*** (0.0545)	-0.459*** (0.116)
Age	0.416*** (0.0193)	0.416*** (0.0194)		
Constant	-2.986*** (0.120)	-2.719*** (0.134)	-1.785*** (0.0925)	-1.614*** (0.135)
Observations	330,104	330,104	293,863	293,863
Other Controls	No	No	Yes	Yes
Log Pseudo Likelihood	-95,142.1	-95,127.9	-84,667.3	-84,662.3
Pseudo R-squared	0.0859	0.0861	0.0892	0.0893

NOTES: The firm-level control variables include the firm's age, subsidies received, export intensity, and financial costs. The city-level control variables include city size, industry composition, and access to foreign capital. All models include province and industry fixed effects. Firm-level and city-level controls are constructed as the control value multiplied with the normalized year. To avoid potential multi-collinearity, we include year fixed effects in specifications without other controls, but not in those with other controls. Standard errors are clustered at the province level and are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Measurement error in the extensive-margin measure could bias the estimates. This issue is unlikely to overturn our main results for two reasons. First, any such error should affect PFs and SOFs similarly. Since our goal is to identify differential effects of banking competition across ownership types, there is little reason to expect such error to bias the estimates systematically toward SOFs. Second, as long as the measurement error is uncorrelated with our key interaction terms, $HHI_{ct} \times SOF_{ict}$ and $HHI_{ct} \times PF_{ict}$, the estimates remain consistent.

To address both measurement-error and endogeneity concerns, we adopt an instrumental-variables strategy to identify the causal effect of banking competition on R&D investment. Following the logic of [Chong et al. \(2013\)](#), we construct an instrument, denoted $HHIiv_{ct}$, as the weighted average HHI of

margin. We appreciate the insight provided by an anonymous referee, which helped clarify this issue for us.

other three cities in the same province with the closest nominal GDP, using those cities' nominal GDP as weights:

$$HHIiv_{ct} = \sum_{s=1}^3 \frac{GDP_{st}}{\sum_{s=1}^3 GDP_{st}} HHI_{st} \quad (27)$$

This instrument strategy is grounded in two key considerations. First, Chinese banks exhibit significant regional segmentation in their deposit and loan activities, primarily to mitigate information asymmetry between lenders and borrowers. Given that most bank loans are secured by collateral, a localized approach is essential for managing financial risks. As a result, enterprises in a particular region often face challenges in obtaining loans from banks located outside their area, thereby reinforcing the characteristics of regional segmentation. In this context, loans from banks in other regions do not directly influence local enterprises' innovation and R&D investments. Second, when banks decide to establish new branches or expand their networks, they typically select locations with GDP sizes similar to those within the same province. These areas tend to attract both similar and diverse types of banks due to comparable strategic motivations. Consequently, the instruments we construct are highly correlated with the endogenous variables and can be considered valid instruments.

We estimate the instrumental variable Probit model (Newey 1987) using the aforementioned instrument, along with its interaction with ownership dummies for the banking competition variable and the associated interaction terms. The estimation results are reported in Table 4. These findings largely align with the baseline Probit model, providing further evidence that intensified banking competition has primarily induced more SOFs to engage in R&D investment.

To test the validity of our instrument, we report a Wald test of the exogeneity of the instrumented variables at the bottom of the table. This test is crucial to ensure that the instrument does not suffer from endogeneity issues, which could bias our results. In the most reliable model presented in the last column, we reject the null hypothesis of no endogeneity at the 10% significance level. This suggests that the instrument is exogenous and that endogeneity is a concern that we are able to control for effectively in this model. However, it is also important to note that when the Wald test statistic is not significant, we cannot reject the null hypothesis of no endogeneity. In such cases, the data may not provide enough evidence to support the use of instrumental variables, and a regular Probit regression might be more appropriate. The point estimates from ivprobit are still consistent, though those from probit are likely to have smaller standard errors. Overall, our estimation results are robust to potential measurement errors in the extensive margin of R&D investment and endogeneity concerns.

TABLE 4
IV-Probit estimates of the extensive margin of R&D investments

Dependent variable:	<i>rdd</i>	<i>rdd</i>	<i>rdd</i>	<i>rdd</i>
HHI	0.0768 (0.135)		-0.0927 (0.218)	
HHI×SOF		-0.316* (0.184)		-0.489** (0.220)
HHI×PF		0.0953 (0.138)		-0.0765 (0.220)
PF	-0.257*** (0.0544)	-0.550*** (0.184)	-0.289*** (0.0571)	-0.581*** (0.181)
Age	0.420*** (0.0182)	0.420*** (0.0182)		
Constant	-3.183*** (0.126)	-2.895*** (0.168)	-1.853*** (0.142)	-1.563*** (0.178)
Observations	297,633	297,633	293,623	293,623
Other Controls	No	No	Yes	Yes
Log Pseudo Likelihood	114,381.3	748,909.0	120,271.3	747,248.4
Wald Test	0.325	3.438	0.182	5.537
Wald Test p-value	0.568	0.179	0.670	0.0627

NOTES: The firm-level control variables include the firm's age, subsidies received, export intensity, and financial costs. The city-level control variables include city size, industry composition, and access to foreign capital. All models include province and industry fixed effects. Firm-level and city-level controls are constructed as the control value multiplied with the normalized year. To avoid potential multi-collinearity, we include year fixed effects in specifications without other controls, but not in those with other controls. Standard errors are clustered at the province level and are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Intensive Margin: Level of R&D Investment

We proceed to test Proposition 3 by investigating the causal impact of banking competition on the intensive margin of innovation investment undertaken by SOFs and PFs. To do this, we focus on the sample of firms that have made positive R&D investment, allowing us to distinguish adjustments in R&D investment along the intensive margin from overall changes in R&D spending.

Our baseline model specification assumes that there is a linear relationship between banking competition and the intensive margin of innovation

TABLE 5
OLS estimates of the intensive margin of R&D investments

Dependent variable:	<i>rdint</i>	<i>rdint</i>	<i>rdint</i>	<i>rdint</i>
HHI	-0.0124*** (0.00392)		-0.0130*** (0.00237)	
HHI×PF		-0.0131*** (0.00406)		-0.0139*** (0.00233)
HHI×SOF		-0.00590 (0.00391)		-0.00251 (0.00459)
PF	-0.000143 (0.00131)	0.00485 (0.00302)	-0.000269 (0.00148)	0.00756** (0.00286)
Age	-0.000805* (0.000445)	-0.000806* (0.000445)		
Constant	0.0263*** (0.00453)	0.0215*** (0.00548)	0.0192*** (0.00294)	0.0114** (0.00425)
Other Controls	No	No	Yes	Yes
Observations	31,560	31,560	28,229	28,229
Adjusted R^2	0.045	0.045	0.057	0.057

NOTES: The firm-level control variables include the firm's age, subsidies received, export intensity, and financial costs. The city-level control variables include city size, industry composition, and access to foreign capital. All models include province and industry fixed effects. Firm-level and city-level controls are constructed as the control value multiplied with the normalized year. To avoid potential multi-collinearity, we include year fixed effects in specifications without other controls, but not in those with other controls. Standard errors are clustered at the province level and are reported in parentheses. Statistical significance is denoted by * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

investment.¹² The linear models are specified as follows:

$$rdint_{ict} = \gamma_0 + \gamma_1 HHI_{ct} + \gamma_2 PF_{ict} + \boldsymbol{\vartheta}_1 \mathbf{Z}'_{ict} + \epsilon_{ict}, \quad (28)$$

$$rdint_{ict} = \lambda_0 + \lambda_1 HHI_{ct} \times PF_{ict} + \lambda_2 HHI_{ct} \times SOF_{ict} + \lambda_3 PF_{ict} + \boldsymbol{\vartheta}_2 \mathbf{Z}'_{ict} + \eta_{ict}, \quad (29)$$

where ϵ_{ict} and η_{ict} are the error terms. We use a set of firm- and city-level controls similar to those in the extensive-margin analysis. The OLS results are reported in Table 5. They point to two conclusions. First, banking competition is positively associated with average R&D intensity. Second, this effect is economically larger and statistically stronger for private firms. This pattern sharply contrasts with the extensive-margin results.

12 We also tried including a quadratic term of HHI_{ct} to see there is non-linear relationship between the level of R&D investment and banking competition; only the linear term is found to be significant and negative.

TABLE 6
2SLS estimates of the intensive margin of R&D investments

Dependent variable:	<i>rdint</i>	<i>rdint</i>	<i>rdint</i>	<i>rdint</i>
HHI	-0.0270*** (0.00273)		-0.0284*** (0.00377)	
HHI×PF		-0.0278*** (0.00248)		-0.0290*** (0.00348)
HHI×SOF		-0.0154 (0.0101)		-0.0168 (0.0119)
PF	-0.000385 (0.00143)	0.00813 (0.00577)	0.000137 (0.00149)	0.00843 (0.00614)
Age	-0.00105** (0.000508)	-0.00105** (0.000507)		
Constant	0.0349*** (0.00460)	0.0262*** (0.00915)	0.0323*** (0.00417)	0.0238** (0.00951)
Other Controls	No	No	Yes	Yes
Observations	28,371	28,371	28,173	28,173
Adjusted R^2	0.048	0.048	0.054	0.054
K-P rk Wald F	271.6	94.86	120.6	61.91
Stock-Yogo Critical Value	16.38	7.03	16.38	7.03

NOTES: Other firm-level control variables include the age, the subsidy the firm receives, the export intensity, and the firm's financial costs. Other city-level control variables include the city size, the industry composition, and the accessibility to foreign capital. All models include province and industry fixed effects. Firm-level and city-level controls are constructed as the control value multiplied with the normalized year. To avoid potential multi-collinearity, we include year fixed effects in specifications without other controls, but not in those with other controls. The Stock-Yogo Critical Value indicate the minimum F-statistic required to avoid weak instruments, assuming that the instruments explain no more than 10% of the endogenous variable's variation. All standard errors are clustered at the province level. Standard errors are in parentheses. Significance denoted by * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

We next turn to an instrumental-variables specification to address the same endogeneity concerns in the intensive-margin regressions. The results, reported in Table 6, show that banking competition has a large and statistically significant effect on the level of R&D investment for private firms, but no significant effect for state-owned firms. Adding the full set of firm- and city-level controls in columns (3) and (4) leaves these conclusions unchanged. The Kleibergen-Paap rk Wald F statistic, reported at the bottom of the table, indicates that the instruments are not weak: in all specifications, the statistic comfortably exceeds the critical value of Stock-Yogo weak identification test.

4.4. Robustness Checks

The baseline empirical results are consistent with the model's main predictions: banking competition has a stronger effect on the probability of

innovation for SOFs and a stronger effect on the intensive margin of R&D investment for PFs. We next examine a series of robustness checks, reported in Online Appendix A.4.

The first concern relates to whether our results are influenced by unobserved firm characteristics that are not accounted for in our econometric model, potentially leading to endogeneity due to omitted variable bias. To address this, we incorporate several firm-level variables that may correlate with banking competition, ownership structure, and innovation activities. Specifically, we examine capital intensity, productivity efficiency, and cash flow. *Capital intensity* may rise with increased banking competition, as competition can lead to lower interest rates on capital rentals, encouraging firms to invest more. Consequently, any observed innovation effects tied to banking competition might be due to spillover effects from new investments. Additionally, capital intensity is often associated with firm ownership, given that SOFs tend to operate in more capital-intensive sectors. *Productivity* also plays a crucial role in determining R&D investment, as firms with varying productivity levels experience different net benefits from innovation. If firms select their locations based on productivity—the selection effect—then banking competition may correlate with firm productivity levels. Studies, such as Guariglia et al. (2011), have indicated that cash flow is vital for Chinese private firms to navigate liquidity constraints and achieve growth. Moreover, *cash flow* is frequently the primary source of R&D financing due to external investors facing asymmetric information challenges (Hall and Lerner 2010). Thus, controlling for cash flow in our estimations is essential. We implement these controls sequentially and then collectively within the model. Our findings consistently indicate that the results remain stable after incorporating these additional firm characteristics.

We then consider the concern that the possibility that the result is driven by the *entry and exit decisions* of firms. To deal with this concern, we construct two variables that characterize the entry and exit behaviors of the firms.¹³ We include them into the estimation of the model (25) to account for the impact of firm entry and exit behavior on the firm's innovation activities. Note that our estimation for analyzing the change at the intensive margin of R&D spending also suffers from the impact of the firm's entry and exit decisions. We tackle this concern by restricting our sample to contain the firms that have consecutive observations and are active in R&D during the sample period. Our results are stable to these tests.

Also, considering that the size of bank loans may vary across different firms, naturally leading to different levels of R&D investment. To account for this size effect that may bias our results for the intensive margin of R&D investment, we include the firm size, measured as the log of total asset, as an

13 A firm entry (exit) is defined when it only appears in year 2007 (2006).

additional control variable. Moreover, we use the ratio of R&D expenditure to total debt as the dependent variable since we do not have information on the total bank loans. Our results remain robust to this alternative specification.

Lastly, we use $CR5$ as the measure of banking concentration. We replace HHI_{ct} with $CR5$ and re-estimate our model with full control of the variables we discussed. Not surprisingly, the estimation results are similar to those we presented in the benchmark analysis.

To the best of our knowledge, both the empirical results and the theoretical framework presented in earlier sections contribute new insights to the existing literature.¹⁴ The observed heterogeneity in default risks and R&D efficiencies across firms underscores the significance of asymmetric information and adverse selection mechanisms, which yield differing impacts on R&D investment among firms with varying ownership structures.

5. Numerical Analysis

In this section, we calibrate the baseline model to study how banking competition affects R&D investment, R&D output, and welfare. We identify the model parameters by matching the model to key empirical moments. Additional calibration details are provided in Online Appendix C.

5.1. Calibration

This section outlines the calibration of model parameters, with one model period corresponding to one year. We begin by calibrating the externally given parameters $\{\kappa_o, b_o, F_o\}_{o \in \{h, l\}}$. For collateral requirements, we assume $\kappa_h = \kappa_l = 1$, implying that all loans are fully collateralized. The return on collateral is normalized to $b_h = b_l = 1$.

For the riskiness distribution, we assume that both LDRFs and HDRFs draw θ from the same distribution $F(\theta)$. We choose a truncated normal distribution $\mathcal{N}((\underline{\theta} + \bar{\theta})/2, 0.02^2)$ on $[\underline{\theta}, \bar{\theta}]$ in the baseline calibration. The numerical CDF, PDF and conditional mean of this distribution are plotted in Figure C1 of Online Appendix C.3. For robustness, Online Appendix C.4 provides numerical results based on alternative distributions, including truncated normal distributions with different parameterization, tuned Beta distributions with various shapes, and a parameterized distribution satisfying Assumption 1. As shown in the appendix, the calibration numerical results are broadly similar under alternative distributions.

The remaining parameters are calibrated to match empirical moments from China data or the literature. Details of parameter identification are provided in Online Appendix C.2, where we show that all parameters are exactly identified. We treat HDRFs as private firms (PFs) and LDRFs as state-owned

14 Refer to the literature section detailing empirical findings in the current body of work.

firms (SOFs), and use sales-weighted empirical moments as calibration targets to account for differences in firm size. Since the model is scale-invariant in firm size, we normalize firms' average sales to 1.

The parameters are chosen as follows. The number of banks is set to $N = 2$, the integer closest to the average bank HHI observed in the data. The share of HDRFs is calibrated to $\lambda_h = 0.804$, matching the empirical share of PFs. The lower bound of the riskiness distribution $\underline{\theta}$ is normalized to 0 since only entrepreneurs with $\theta \geq \hat{\theta}_o$ obtain bank loans. The upper bound of the riskiness distribution is set to $\bar{\theta} = 0.189$ to match the non-performing loan (NPL) share of PFs, which is 11.7%. For the subsidy probability, although the observed NPL share of SOFs is non-zero, we consider banks facing the defaulted loans can be subsidized in an alternative approach such that the effective NPL share of SOFs is zero. This implies a subsidy probability $\delta = 1$. The empirical NPL shares are calculated using 2009 data on NPLs by firm type from the Chinese Banking System Database compiled by the China Stock Market & Accounting Research Database (CSMAR), the earliest year with available observations. Details of the construction are provided in Online Appendix C.1. The collateral valuation cost is calibrated to $\gamma = 0.020$ to match the extensive margin of PFs at 0.313, consistent with empirical estimates of bankruptcy costs in [Bris et al. \(2006\)](#). We set LDRFs' R&D investment limit $\bar{z}_l$ to 0.008 to match SOFs' intensive margin. For HDRFs, we assume a credit-rationing equilibrium in which the firms' R&D investment constraint is nonbinding. We set HDRFs' R&D investment limit, $\bar{z}_h$, to 0.015, which exceeds PFs' intensive margin of 0.010. This calibration allows us to explore counterfactual scenarios in which the HDRFs' R&D investment constraint becomes binding. The unit R&D productivity parameters μ_h and μ_l are calibrated to match the loan rates from [Li et al. \(2024\)](#): 5.75% for PFs and 4.48% for SOFs, implying $\mu_h = 1.052$ and $\mu_l = 1.041$. We set the return on liquid assets to $\rho_r = 2.80\%$, consistent with the average values of SHIBOR and Treasury yields over 2006 to 2010.¹⁵ For the deposit market, we choose the deposit rate $\bar{r}_d = 2.40\%$ to match the average deposit rate in China over 2000 to 2010. The deposit supply $\bar{D}$ is set to 0.0038 that matches a net interest margin of 2.46%, consistent with the empirical values over 2000 to 2010.

We assume banks' capacity cost functions are quadratic:

$$c_h(L_h) = \frac{\chi_h}{2} L_h^2, \quad c_l(L_l) = \frac{\chi_l}{2} L_l^2, \quad c_r(R) = \frac{\chi_r}{2} R^2, \quad (30)$$

where χ_o captures the convexity of the cost functions. The parameter χ_h is set to 17.955 to match PFs' intensive margin of 0.010, and χ_l is set to 33.197 to match SOFs' extensive margin of 0.675. The parameter χ_r is set

15 We choose this time period because the SHIBOR rate is publicly available starting in October 2006, and the data periods for the empirical loan interest rate in [Li et al. \(2024\)](#) is from March 2006 to June 2010.

to 81.541 to match a cost-to-income ratio of 0.5, consistent with the average value of Chinese commercial banks over 2000 to 2010. The full set of calibrated parameters, along with their targets or sources, is summarized in Table 7.

Given the calibration, the parameter values satisfy the properties needed for the equilibrium characterization. As shown in Figure C2 of Online Appendix C.3, the unit return on HDRF loans is hump-shaped with a unique interior global maximum, whereas the unit return on LDRF loans is monotonically increasing in r_l .¹⁶ Moreover, the monopoly revenue functions are both hump-shaped with unique interior global maxima. The implied return-maximizing rates are $\hat{r}_h = 5.75\%$ and $\hat{r}_l = 5.05\%$, while monopoly revenue in the second stage is maximized at $r_h^m = 5.28\%$ and $r_l^m = 4.20\%$ for HDRFs and LDRFs, respectively. For the threshold values of N in Proposition 2, our calibration implies $\bar{N}_l = 58.156$, $\bar{N}_h = 73.939$ and $N_h^r = 7.921$. Moreover, as shown in Online Appendix C.3, Figure C4 numerically verifies the uniqueness of equilibrium by showing that there is a unique solution to equilibrium condition (13) and (14), and Figure C5 numerically shows the uniqueness of N_h^r and $\bar{N}_l$.

5.2. Counterfactual Analysis

Figure 1 presents the comparative statics of bank competition under the baseline scenario. The horizontal axis in each panel shows the competition index—defined as $1 - \text{bank HHI}$ in the theoretical model—ranging from 0 to 0.967, which spans the empirical range of bank concentration documented in Table 1.

The numerical analysis serves two purposes. It shows how the model's mechanisms operate over a wider range of competition levels than the reduced-form evidence can capture, and it clarifies which results change when the equilibrium shifts between credit-rationing and market-clearing regimes. We therefore focus on the economically meaningful objects: loan allocation across firm types, the two margins of R&D investment, aggregate loan risk, and welfare.

When the competition index is below 0.875, the HDRF loan market is credit rationed, and the comparative statics are consistent with model predictions in Section 3.5. As competition intensifies, the loan rate for SOFs declines while the loan rate for PFs remains unchanged. At the bank level, loans per bank fall for both groups, whereas aggregate lending rises for both groups. Aggregate lending rises more strongly for PFs than for SOFs, so the SOF-to-PF loan ratio declines as competition increases.

R&D investment exhibits heterogeneous responses across firm types. Under a larger competition index, the extensive margin increases for SOFs but

16 Although the HDRF loan return increases at the right boundary, all the theoretical results remain valid, as the global maximum is interior and the return is strictly increasing to the left of this maximum. See the proof of Proposition 1 for details.

TABLE 7
Model calibration

Parameters	Description	Value	Source or target
<i>A. Externally calibrated parameters</i>			
κ_h, κ_l	Collateral ratio	1	Full collateralization
b_h, b_l	Return on collateral assets	1	Normalization
F_h, F_l	R&D riskiness distribution	$\mathcal{N}(\frac{\theta+\bar{\theta}}{2}, 0.02^2)$	Truncated on $[\underline{\theta}, \bar{\theta}]$
$\underline{\theta}$	Lower bound of riskiness type	0	Normalization
$\bar{z}_h$	HDRFs R&D investment limit	0.015	Higher than PFs' intensive margin
<i>B. Parameters from matching moments</i>			
N	Number of banks	2	Average bank HHI=0.7
λ_h	Share of HDRFs	0.804	Share of PFs=0.804
$\bar{\theta}$	Upper bound of riskiness type	0.189	PFs' NPL share=11.7%
γ	Collateral valuation cost	0.020	PFs' extensive margin=0.313
δ	Probability of subsidy	1	SOFs' NPL share=0
χ_h	HDRF capacity cost	17.955	PFs' intensive margin = 0.010
χ_l	LDRF capacity cost	33.197	SOFs' extensive margin = 0.675
χ_r	Reserve management cost	81.541	Cost-income ratio = 0.5
$\bar{D}$	Deposit supply	0.0038	Net interest margin = 2.46%
$\bar{z}_l$	LDRFs R&D investment limit	0.008	SOFs' intensive margin=0.008
μ_h	HDRF R&D productivity	1.052	PFs' loan rate=5.75%
μ_l	LDRF R&D productivity	1.041	SOFs' loan rate=4.48%
ρ_r	Return on liquid assets	2.80%	Mean of treasury yield and SHIBOR
$\bar{r}_d$	Deposit rate	2.40%	Average deposit rate=2.40%

remains unchanged for PFs. In contrast, the intensive margin is constant for SOFs but rises for PFs. Turning to loan risk, the aggregate NPL share increases with the competition index, indicating that intensified bank competition generates a reallocation of loans towards PFs, which are inherently riskier, and thereby drives up the aggregate NPL share. Finally, the unit loan return declines for SOFs but remains unchanged for PFs under higher competition index.

When the competition index exceeds 0.875, the HDRF market transitions to a market-clearing equilibrium, with loan quantities determined by Cournot capacities. In this regime, loan rates for both HDRFs and LDRFs decline with bank competition index. The qualitative patterns in quantities persist: loans per bank decrease, aggregate lending increases, and the SOF-to-PF loan ratio continues to fall.

The behavior of R&D margins changes. For both firm types, the extensive margins increase with the competition index, while intensive margins are constant. In terms of loan risk, the aggregate NPL share becomes hump-shaped in the competition index. This is because the decline in the HDRF loan rate induces entry of PFs with lower default risk, which dominates when

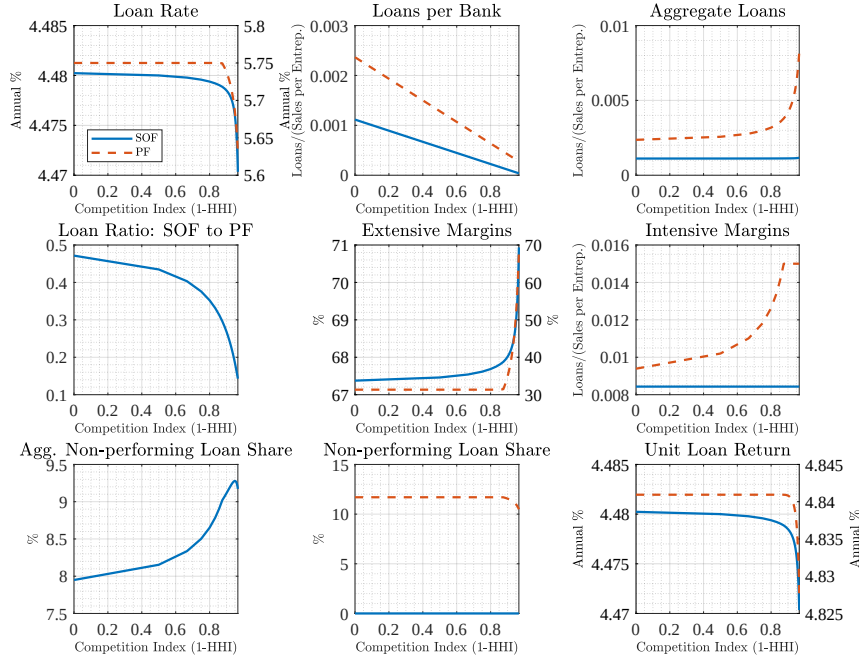


FIGURE 1 Comparative statics of bank competition in the baseline scenario

NOTES: The horizontal axis shows the competition index ($1 - \text{bank HHI}$). In panels with dual vertical axes, the left axis corresponds to the blue solid line (SOF), and the right axis corresponds to the red dashed line (PF). The parameter values are reported in Table 7.

the competition index is sufficiently large. Finally, the unit loan returns decline for both firm types as bank competition intensifies.

Note that the increase in aggregate loan risk under increased competition does not reflect weaker screening or worsening adverse selection. Instead, it arises from an endogenous shift in banks' loan portfolio toward private firms with higher default risk. Within the SOF loan market, the loan quality actually improves with competition, because lower loan rate for SOFs induce entry by SOFs with lower default risk (i.e., those with lower θ_l). This distinction between within-segment loan risk and aggregate portfolio risk is central to the mechanism.

From a policy perspective, our results suggest that an observed increase in aggregate loan risk following stronger bank competition need not signal deteriorating screening standards. Instead, it may reflect a socially relevant reallocation of credit toward underserved but riskier private firms. This distinction is important for interpreting regulatory responses to rising non-

performing loans in competitive banking systems. In our framework, the pattern is driven by adverse selection in a multi-market lending environment together with implicit guarantees in the SOF market.

Next, we conduct a counterfactual analysis to explore how government subsidies for SOFs' loan default affects the impact of bank competition. In the baseline calibration, $\mu_l < \mu_h$, indicating that SOFs are less efficient in R&D than PFs. However, due to government loan subsidies, SOFs are still fully financed when they receive bank loans, and their extensive margin is higher than PFs. Thus, the primary source of loan misallocation arises from this policy support.

To isolate this effect, we eliminate the government subsidy by setting $\delta = 0$. In this scenario, SOFs and PFs differ only in R&D efficiency and loan capacity costs: SOF R&D is less productive, and SOF loan capacity is more costly. As shown in Online Appendix C.3, Figure C3 plots the unit loan return and monopoly revenue of SOF loans. The unit loan return is maximized at $\hat{r}_l = 4.55\%$, the monopoly loan revenue is maximized at $r_l^m = 4.20\%$, and SOF loan returns are uniformly below those of PFs (see the top panels of Figure C2). The implied threshold is $\bar{N}_l = 100.668$. Figure C6 plots the equilibrium individual LDRF loan capacity L_l^* as function of N . There exists a second threshold, $N_l^r = 3.336$, such that $L_l^* \geq \hat{L}_l/N$ for $N < N_l^r$, and $L_l^* < \hat{L}_l/N$ for $N > N_l^r$. Accordingly, the LDRF market exhibits a credit-rationing equilibrium when the competition index is above 0.70, and a market-clearing equilibrium when the index is below 0.70.

Figure 2 compares the effects of bank competition in the baseline and in the counterfactual without government loan guarantees. Because the equilibrium outcomes for PFs are qualitatively and quantitatively similar across the two scenarios, we focus on the SOF market.

We first observe that, compared to the baseline, eliminating loan guarantees substantially reduces the unit return on SOF loans due to higher default risk. This weakens banks' incentives to lend to SOFs, leading them to reduce SOF loan supply and charge higher loan rates. As a consequence, SOF lending, as well as the extensive and intensive margins of SOF R&D investment, are lower in the counterfactual, and banks reallocate more credit toward PFs, as reflected in a lower SOF-to-PF loan ratio. At the same time, heightened SOF risk and the shift in loan composition raise the aggregate NPL share. Moreover, as bank competition intensifies, the contraction in SOF lending becomes more pronounced and drives the market deeper into credit rationing. In particular, aggregate SOF loans and both innovation margins decline with the competition index; once the index exceeds 0.89, SOF loan supply falls to zero. These results suggest that, under our calibration, the curvature $c_r'' = \chi_r$ is sufficiently large, implying strong substitution effects between SOF and PF loans and a pronounced deposit-competition effect, both of which significantly compress SOF loan supply.

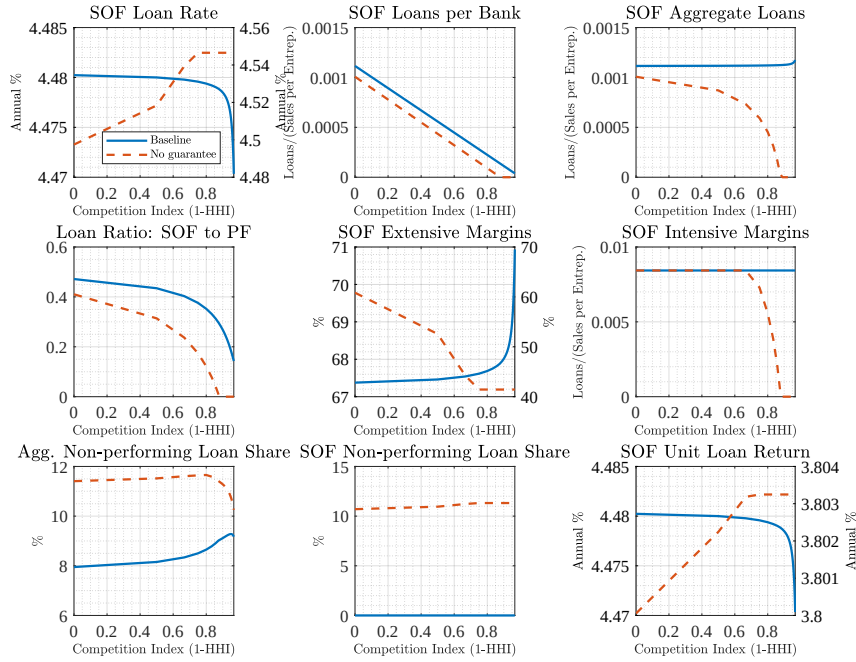


FIGURE 2 Counterfactual analysis: removing loan guarantee for SOFs

NOTES: The horizontal axis represents the competition index ($1 - \text{HHI}$). In panels with dual vertical axes, the left axis corresponds to the blue solid line (Baseline), and the right axis corresponds to the red dashed line (No guarantee). We set $\delta = 0$ in the “No guarantee” case; all remaining parameter values for both cases are reported in Table 7.

Figure 3 compares aggregate R&D output and welfare across the baseline and counterfactual scenarios. R&D output is defined in equation (18), bank welfare in (21), and welfare for SOFs and PFs in (19) and (20), respectively. Removing loan guarantees reallocates credit toward PFs and thereby raises aggregate R&D output, with a peak gain of 0.54% relative to the baseline at a competition index of 0.875. At the same time, bank welfare is lower in the counterfactual because returns on SOF loans decline. PF welfare rises as loan supply expands, whereas SOF welfare falls as loan supply contracts. Moreover, in the baseline, stronger bank competition raises both R&D output and aggregate welfare for all agents. In contrast, in the counterfactual, SOF welfare declines with the competition index and eventually falls to zero as loan supply contracts.

In sum, the counterfactual analysis shows that removing loan guarantees reduces SOFs’ access to credit, thereby lowering their R&D investment and

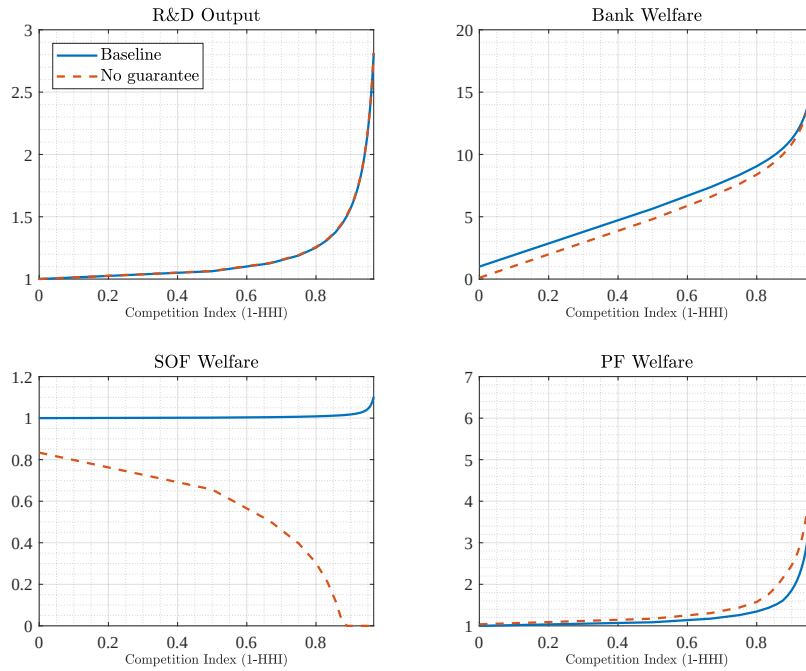


FIGURE 3 Comparison of R&D output and welfare: baseline vs. counterfactual
 NOTES: In all panels, variables are normalized by their baseline values at competition index = 0. The top panels plot aggregate R&D output and aggregate bank welfare, while the bottom panels show aggregate welfare of SOFs and PFs. The horizontal axis represents the competition index ($1 - \text{bank HHI}$). We set $\delta = 0$ in the “No guarantee” case; all remaining parameter values for both cases are reported in Table 7.

welfare. At the same time, it increases overall R&D output and enhances PFs’ R&D investment and welfare through credit reallocation. These findings suggest that adverse selection in the loan market can be sufficiently strong that loan guarantees to SOFs distort bank credit allocation and ultimately hinder aggregate innovation.

6. Conclusion

This paper studies how bank competition affects innovation when firms differ in default risk and credit access. The main message is that the effect of competition is heterogeneous not only across firms but also across innovation margins. In our model, low-default-risk firms are fully funded, so more competition mainly lowers loan rates and brings additional firms into R&D. High-default-risk firms are credit rationed, so more competition

mainly increases individual loan size and raises R&D investment conditional on innovation. The same framework also implies that aggregate loan risk can rise even when within-segment loan quality improves, because competition reallocates credit toward riskier borrowers.

We test these mechanisms using Chinese firm-level and city-level banking data around the deregulation of city commercial banks. The empirical evidence mirrors the model's central predictions: stronger banking competition increases the probability that SOFs engage in R&D, but primarily raises the intensity of R&D investment for PFs. The quantitative analysis extends this logic and shows that the aggregate NPL share can be hump-shaped once the high-default-risk loan market shifts from credit rationing to market clearing. It also shows that removing loan guarantees for SOFs reallocates credit toward PFs, increases aggregate R&D output, and lowers the welfare of banks and SOFs.

Taken together, the results suggest that the relationship between bank competition, innovation, and financial stability is fundamentally about composition. Competition does not simply expand or contract innovation uniformly. It changes which firms receive credit, through which margin they respond, and how bank portfolios are reallocated across safer and riskier borrowers. That perspective helps explain why policies that protect incumbent low-risk borrowers may stabilize one segment of the credit market while distorting the allocation of innovative activity in the economy.

Supporting information

Additional supporting information can be found in the online version of this article.

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